Two-Facet Corners for Large Weight-Driven Pipe Organ Reservoirs: Adaptation and Variation

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Abstract: Two-facet corners are designed so that as much of the reservoir corner as feasible is constructed from rigid material, rather than flexible leather or rubber-cloth, to maximize the corner's durability, maintainability, and precision of pressure regulation. The geometry and parametrical error properties of the two-facet outward-folding corner (first introduced in the context of voided reservoirs) are explored. A newly-discovered two-facet alternate-folding corner type is introduced that has analogous geometric properties to the two-facet outward corner. Banausic fabrication algorithms for the rib fragments of two-facet corners are presented, that facilitate their use in reservoirs large enough that the weight of the rib material would strain traditional cloth and leather inter-rib hinges. A novel type of single-level voided reservoir is described, enabled by the two-facet alternate-folding corner. An imperfect type of camera-folded reservoir, enabled by the commensurable species of two-facet alternate-folding corner is also described. Thought-experiments in designing very large reservoirs are presented.

1. Background and Significance

In the early 2000's the author asked the question on the old MMD Pipes Forum as to how to build a weight-driven air pressure reservoir that would have equal output pressure at all reservoir inflation heights, with a view towards avoiding the complexity and pressure imprecision of conventional coiled reservoir springs. At the time, the only known methods were to build a 'camera-folded' reservoir (two inward and two outward ribs with rubber-cloth or leather covering the gaps at the corners) or a classic 'double-rise' linked inward and outward reservoir. These types are described and illustrated at [1]. But both are problematic: the 'camera folded' reservoir requires a big unsupported piece of rubber-cloth or leather which undergoes kinking and rolling motions, which cause overly rapid wear, and the 'double-rise' reservoir requires a specialized hinged linkage that introduces a (small) amount of friction and requires a decent internal bearing.

Thus, the author began to wonder how to build a weight-driven air pressure reservoir that would have exactly equal pressure at all reservoir inflation heights, with rigid corner structures moving only by low-friction and low-wear folding movements of the traditional canvas hinges, which could be built with either leather or (less elastic) rubber-cloth sealing strips, with no need to purchase or fabricate springs or linkages.

By the mid 2000's, with his interest in organ-building waning, the author had not yet found the answer. But in January 2018, the idea of a reservoir with internal compensating voids, built around the Aeolian-style inward-folding corners as described at [3] presented itself to the author's mind. That initial concept turned out to be too capacity-inefficient, but early in March 2025 the author realized that a reservoir with internal compensating voids could be made capacity-efficient by using outward-folding ribs; the geometry of the two-facet outward-folding corner presented itself starting in late March 2025. The design of this voided reservoir is presented in [2].

Although the voided reservoir may well be one of the most elegant discoveries in the history of organ wind supply, its description in [2] poses two practical issues: First, for extremely large reservoirs, the weight of the ribs and rib fragments (whose weight may increase as the square of the reservoir height) places unacceptably high strain on the cloth hinges joining them, the hinges' size and strength only increasing, at most, linearly with reservoir height. This problem is especially acute with regard to the triangular rib

fragment of the two-facet corner. Second, due to the long void space needing to be located below the main reservoir chamber, an element of vertical spatial inefficiency is present.

2. Brief Definitions

The lateral surfaces (*rib-band*) of a main reservoir chamber or a void space is composed of *rib pairs* and *rib fragment pairs*, which form dihedral angles that change as the chamber's or void's lid rises and falls due to air outflow, inflow, and inflation level. A corner of a rib-band where two rib pairs join is comprised of *rib fragments* which are generally triangular, as compared to the trapezoidal shape of the ribs themselves. The rib pairs and corners combine to delineate an interior, finite region and an exterior, infinite region. A rib pair is said to be *outward-folding* if the line connecting them is always further from the center of the interior region at every permissible inflation level of the interior region than the centroids of the ribs in the pair - otherwise the rib pair is said to be *inward-folding*. A two-facet corner can likewise be considered inward-folding, similarly depending on the location of the line joining the two rib fragments or facets. A two-facet corner is said to be an *outward-folding corner* if both rib pairs it connects are outward-folding; an *inward-folding corner* if both rib pairs it connects are inward-folding; and an *alternate-folding corner* if one rib pair is outward-folding and the other rib pair is inward-folding.

3. Review of the Design of the Two-Facet Outward-Folding Corner

One deficiency of traditional reservoir corner construction is that much of the corner consists of a flexible gusset (e.g. leather) allowing the corner geometry to vary quite a bit when inflated; this large gusset can bulge out significantly and comically. In addition, the gusset's notable expansion and compression causes wear it over time. This is especially true when one wishes to avoid the use of leather, as rubber-cloth is less supple than leather. For small reservoirs and regulators this may be a minor issue, but as reservoir size increases, the drawbacks of traditional corner construction become much more apparent.

To allow essentially the entire rib-band of a reservoir's rib-set(s) to be rigid, and able to fold flat or conform to any allowable inflation height (where the maximum fully inflated height corresponds to a 90° dihedral angle between ribs in a rib pair forming one side of a reservoir), a two-facet outward-folding corner pattern, constructed of two triangular rib fragments per corner, may be used.

Two outward-folding rib pairs meet to form an outward corner. Figure 1 shows them seen from above (i.e. in plan view) completely flat.

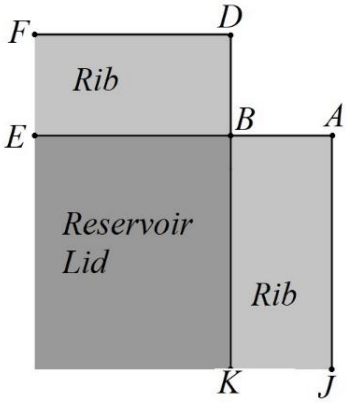


Figure 1

For the analyses presented in this paper, all lengths should be understood in terms of being multiples of rib width; that is to say, rib width (i.e. the length *DB* and *AB* of Figure 1) is taken as the standard unit of measurement. Similarly, areas are considered multiples of a unit of area defined as 1 square rib width. Said another way, in comparison to all the lengths presented herein, the rib width of a given rib-band is

considered to have relative measure of 1. Thus, to find the length in real-world units of a relative length mentioned in this paper, one merely has to multiply by the real-world rib width of the reservoir being designed.

Figure 2 illustrates the process of developing a two-facet outward corner: From the rib configuration of Figure 1, we extend one rib pair to form a sort of beak, and trim the other rib pair. These changes are in accordance with a corner compactness factor, called the *a* value, which being a length, is a positive real number, that is, $a \in \mathbb{R}^+$. The *a* value should be considered as a multiple relative to the rib width. The length *DG* is the rib width times *a*.

Having chosen the *a* value, the rest of the corner geometry can be inferred. Another value, *d*, is important. The *d* value, being non-negative, defines the length of *CH*. Optimally[1]:

$$d = a - \frac{1}{\sqrt{2}}$$

The rightmost diagram of Figure 2 shows how a rib fragment is constructed as a triangle formed by points *G*, *B*, and *H* . In this example diagram, *a* = 3, and *d* is then $3 - \frac{1}{\sqrt{2}}$

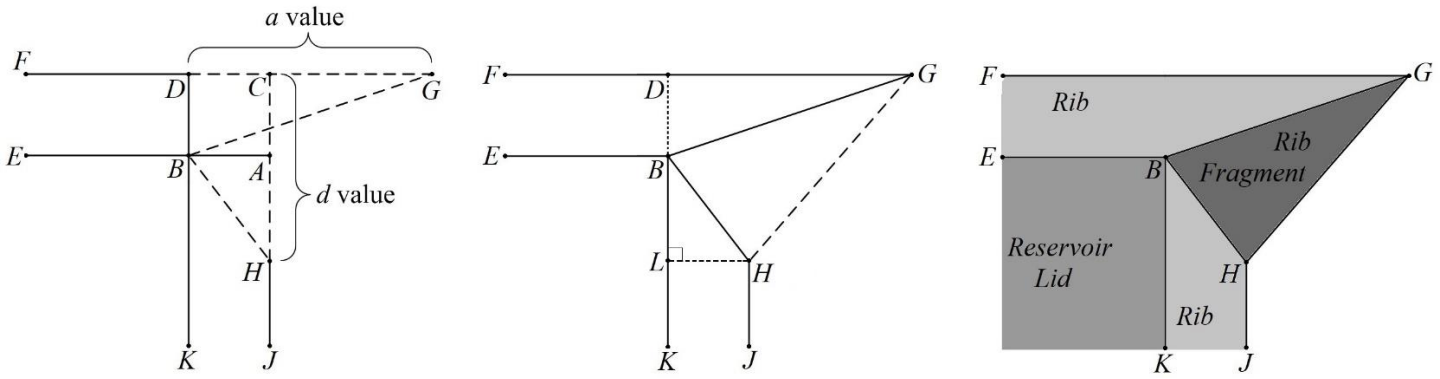


Figure 2

(machine-readable description of Figure 2: A horizontal line segment FG with points C and D intermediately between, points are ordered from left to right as F, D, C, G. Distances: DC=1, DG=3 A horizontal line segment EA with point B intermediately between, points are ordered from left to right as E, B, A. Distance BA=1 A vertical line segment DK with points B and L intermediately between, points are ordered from top to bottom as D, B, L, K. Distances DB=1, BL≈1.29 A vertical line segment CJ with points A and H intermediately between, points are ordered from top to bottom as C, A, H, J. Distances CA=1, AH≈1.29 Additional segments drawn: from L to H, from H to G, from B to H, and from B to G. Point B is common to segment DK and segment EA. Point A is common to segment EA and segment CJ.)

In essence, the idea is to add two triangular outward-folding rib fragments to close the corner.

The two-facet corner has a permissible inflation range from 0° to 90°, measured as the dihedral angle between the ribs of each rib pair. The perspective drawings of Figure 3 show a two-facet corner at its maximum permissible inflation of 90°.

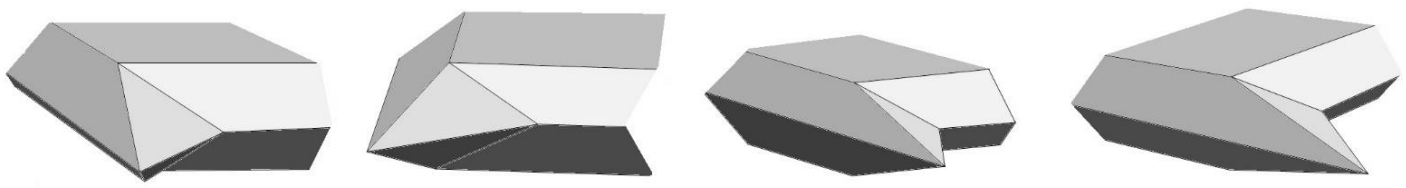


Figure 3

[1] A proof that this *d* value is optimal for all $a > 1$ is sketched in Section 18.

In general, *a* values less than 2, although they produce compact corners, result in a corner geometry quite sensitive to machining variance and with large parametrical error (defined below) over the allowable range of inflation heights. In particular, *a* values in the range below 1.6 give unacceptably large parametrical errors even for a rib width as small as 6 inches.

A special value of *a*, which may be notated as a_c (c: commensurable) being approximately[2] 2.0517… was initially found numerically that seemed to give $\triangle BGH$ an altitude of 1 and thus the same dihedral angle between the two *BGH* rib fragments as the inter-rib dihedral angle over the range 0° and 90°. We can analogously notate: $d_c = a_c - \frac{1}{\sqrt{2}}$

Section 15 presents a proof that

$$\frac{2+\sqrt{2}+\sqrt{6+12\sqrt{2}}}{4}$$

exactly expresses a_c along with a couple of other elegant results.

In section 13 below, we derive a remarkable result: when constructed with an *a* value of a_c , a two-facet corner (i.e. the trapezoid *DGHL*) has an area of a_c. Thus, an outward-folding corner constructed with an *a* value of a_c and *d* value of d_c is equivalent in terms of air pressure effect to a piece of outward-folding lateral rib face of width 1 and length a_c .

For similar rationales as in [2], this paper will mainly focus on *a* values of at least a_c .

At exactly a_c the *HG* dihedral equals the inter-rib angle for all allowable inflation heights; for *a* values above a_c the *HG* dihedral is less than the inter-rib angle for all allowable inflation heights. Corners with an *a* value of a_c are especially useful, however, if their comparatively large parametrical error can be tolerated, since they allow immediate commensurability between the rib fragments and the ribs.

In general, the larger the *a* value, the more tolerant the corner geometry is to machining variance and the smaller the parametrical error (described in section 5 below) over the allowable range of inflation heights. Figure 4 shows the corner geometry (seen from above in plan view when the corner is fully deflated) for a variety of *a* values including a_c.

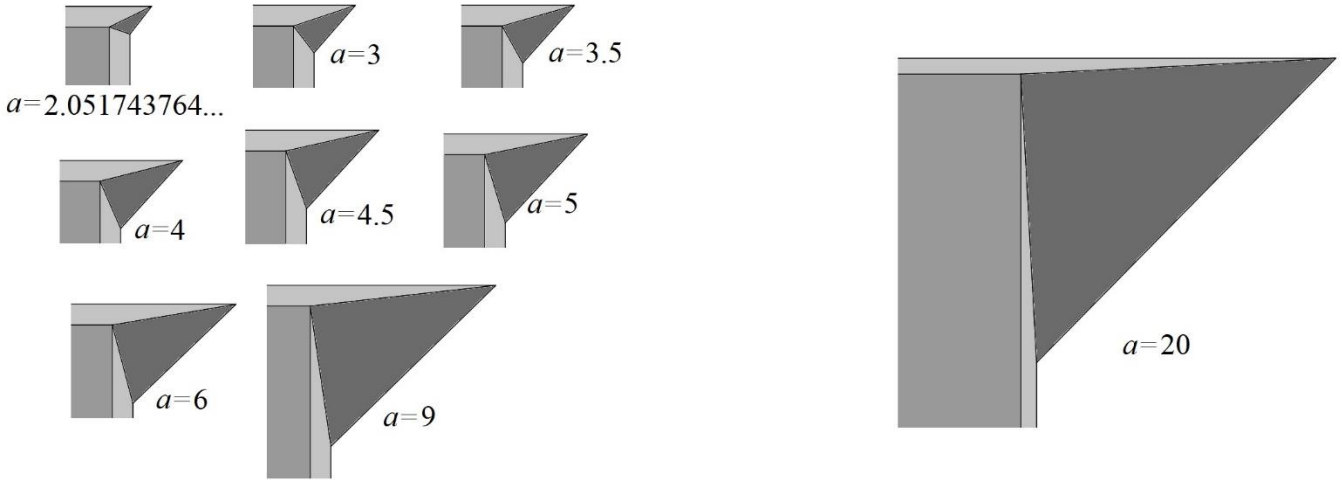


Figure 4

[2] 2.0517437643569012283585648304284633061279537067…

In a voided reservoir (as described in [2]) the *a* value characterizing the corners of the void must match the *a* value characterizing the upper main reservoir chamber -- one cannot mix and match *a* values within the same reservoir.

The dimensions of the ribs and rib fragments will normally initially be calculated exactly and measured out for machining; the best precision possible in measuring and machining is desirable.

Lemma 1. Taking the rib width as 1 (the rib width defines 1 unit) the relative lengths of other segments are:

segments	**Length**
BG	$\sqrt{1+a^2}$
BH	$\sqrt{1+(d-1)^2}$
GH	$\sqrt{d^2+(a-1)^2}$

Proof. Follow immediately from the Pythagorean theorem. ■

Lemma 2. Some key angle measures are:

angles	**measure**
$\angle ABG$ and $\angle BGC$	$\arctan\frac{1}{a}$
$\angle KBH$ and $\angle BHC$	$\arctan\frac{1}{(d-1)}$
$\angle HBA$	$\arctan(d-1)$
$\angle HBG$	$\angle HBA + \angle ABG$
$\angle GHC$	$\arctan\frac{(a-1)}{d}$
$\angle GHB$	$\angle BHC + \angle GHC$
$\angle HGB$	$180° - \angle GHB - \angle HBG$

Proof. By definition of arctangent. ■

A general conception of an outward-folding rib pair is depicted in elevation view Figure 5, where the thickness of the ribs, height of the support nubs, and size of the hinges are greatly exaggerated for clarity.

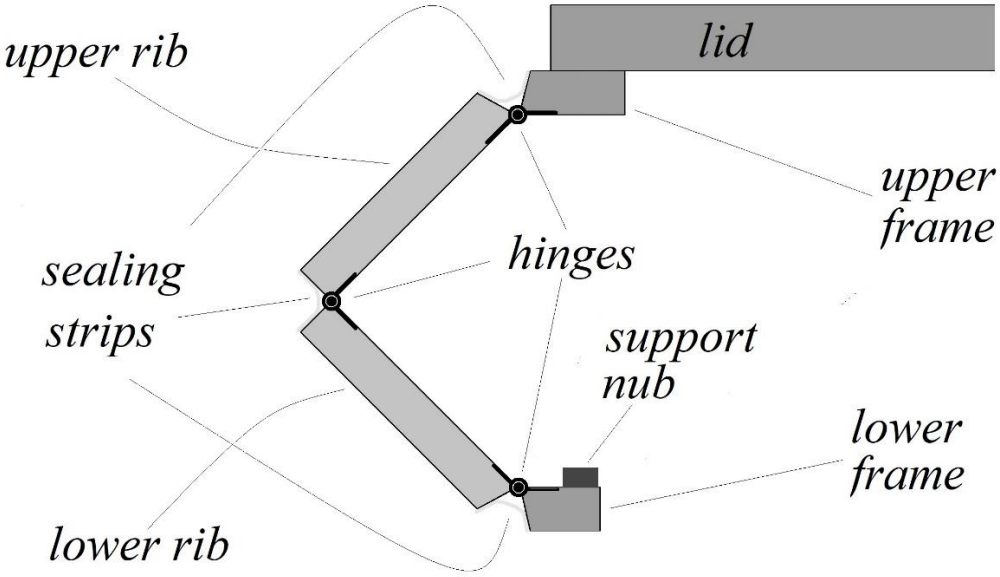


Figure 5

In general, to minimize the area of the sealing strips (e.g. made of leather or rubber-cloth) that is exposed to pressurized air, and to minimize the height of the support nubs, it seems that using many smaller hinges is preferable to using a few larger hinges. In a voided reservoir, the diameter of the hinge cores should be identical in the main reservoir chamber and void space. The recesses to accommodate the hinge plates and hinge centers should be machined into the ribs, rib fragments, and frame pieces prior to setup and assembly. The center-lines of the hinges' cores should be placed to exactly coincide with the dihedral line of contact between the inside surfaces of the ribs, rib fragments, and frames. All screws and hinge plates should be flush with the inside surfaces of all ribs, rib fragments, and frames, so that only hinge cores protrude beyond the lower rib and lower frame surface, and upper rib and upper frame surface, respectively. To prevent the weight of the upper portions of the reservoir or void space from resting on the hinge cores, support nubs strong enough to bear the full weight of the upper ribs, rib fragments, and lid, and having a height slightly more than the hinge cores' diameter, should be installed. All these features are illustrated in Figure 5.

The hinges should be made of brass, stainless steel, or other material that will not rust when water is applied in the vicinity (to soften the glue attaching the sealing strips) when the sealing strips need to be replaced during renovations of the reservoir.

Rather than attaching the ribs and rib fragments directly to the reservoir (or void space) lid and trunk-band, it may be more convenient to attach them to a rectangular frame. The joints between the upper frame and upper ribs, and between the lower frame, and lower ribs, may be beveled a bit as shown in Figure 5. The joints between the ribs and rib fragments (i.e. along the lines *BG* and *BH* of Figure 2) should also be beveled. This beveling will help provide more glued contact surface for the sealing strips.

Enough recessing should be machined along the top and bottom edges of the ribs (i.e. along lines *BE* and *BK* of Figure 2) so that the thickness of the sealing strips does not wedge the frame and ribs apart at all, which would prevent the reservoir from laying as fully flat as possible when deflated. The edges where rib pairs and fragments pairs join vertically (i.e. along lines *HJ*, *GF*, *HV'*, *GV''*, *HV*, *GP''*, etc. of Figure 12 through Figure 17) likely do not need recessing for their sealing strips but perhaps only beveling to increase the glued contact surface.

4. Overview and Design of the Two-Facet Alternate-Folding Corner

Section 3 of this paper provided much of the geometry of the two-facet outward-folding corner. Here we turn our attention to another corner type, the two-facet alternate-folding corner, the development of which is illustrated in Figure 6.

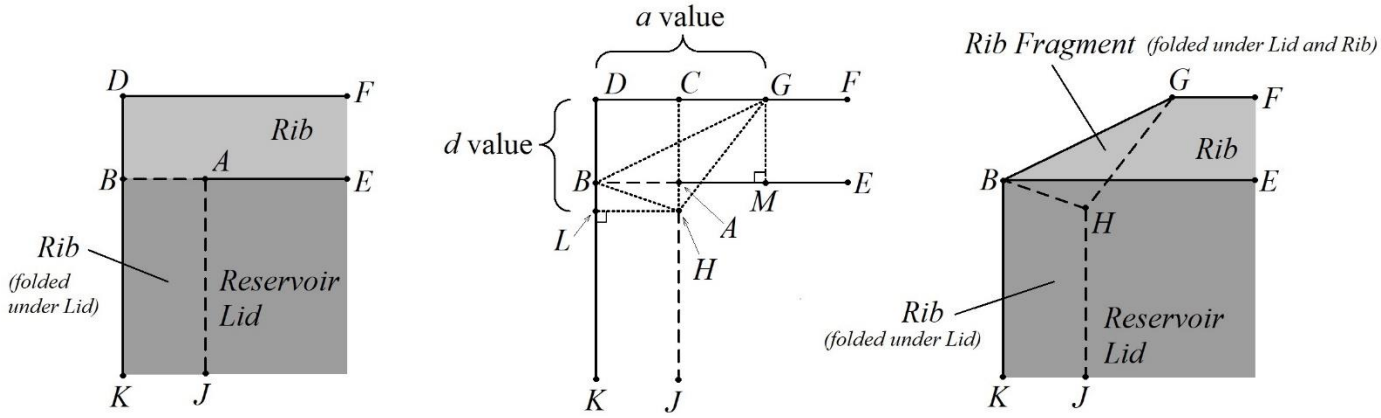


Figure 6

(machine-readable description of Figure 6: A horizontal line segment DF with points C and G intermediately between, points are ordered from left to right as D, C, G, F. Distances: DC=1, DG≈2.0517 A horizontal line segment BE with points A and M intermediately between, points are ordered from left to right as B, A, M, E. Distances BA=1, BM≈2.0517 A vertical line segment DK with points B and L intermediately between, points are ordered from top to bottom as D, B, L, K. Distances DB=1, BL≈0.34 A vertical line segment CJ with points A and H intermediately between, points are ordered from top to bottom as C, A, H, J. Distances CA=1, AH≈0.34 Additional segments drawn: from L to H, from G to M, from H to G, from B to H, and from B to G.)

It will be appreciated that, due to the very analogous nature of the two-facet alternate-folding corner, the same point labeling conventions are used here as in Figure 1 and Figure 2. Likewise, everything explained regarding parametrical error and the trimming of the gusset-gap in the rib fragments in sections 3 through 7 above applies equally well to the two-facet alternate-folding corner.

At the left of Figure 6, we can see the initial set-up of a reservoir corner in plan view in its fully collapsed state. As in Figure 1, there is a reservoir lid (outlined by points *K*, *B*, and *E*) and an outward-folding rib pair (outlined as ▭*DFEB*). The adjacent rib pair (outlined as ▭*BKJA* along the reservoir's *BK* side) is folded inward, as indicated by the dashed lines showing that it's hidden, tucked under the reservoir lid. We must find a way to cover the gap left at the corner.

The central diagram of Figure 6 shows that geometrically constructing the rib fragment to cover the gap can be done in an exactly analogous way to the configuration of Figure 2. We extend *JA* to intersect *DF* at point *C*, choose an *a* value ($a = a_c$ for this diagram, but any positive real number *a* value works) and locate point *G* along *DF* such that *DG* equals that *a* value. We also select a *d* value (optimally $d = a - \frac{1}{\sqrt{2}}$ as noted in Section 3) and locate a point *H* along *CJ* so that the distance *CH* equals the *d* value. Perpendicular lines can be dropped from *H* to segment *KB* to locate point *L*, and from *G* to segment *AE* to locate point *M*. Finally, connect *BG*, *GH*, and *HB*.

The right-side diagram of Figure 6 shows the plan view of the completed corner. Here the outward-folding rib pair *EBGF* and the inward-folding rib pair *JHBK* are shown with their acute-angled distal ends. The rib fragments *BHG* are shown also. It's notable that, unlike the outward-folding corner, the alternate-folding corner requires that the rib fragments be tucked under the reservoir lid and partially between the outward-folding rib pair. Accordingly, the vertical arrangement of ribs and rib fragments is not as simple as in the outward-folding corner.

Figure 7 shows a perspective sketch of the two-facet alternate-folding corner along with a labeled diagram relating the perspective sketch to the plan view of the collapsed corner shown in Figure 6.

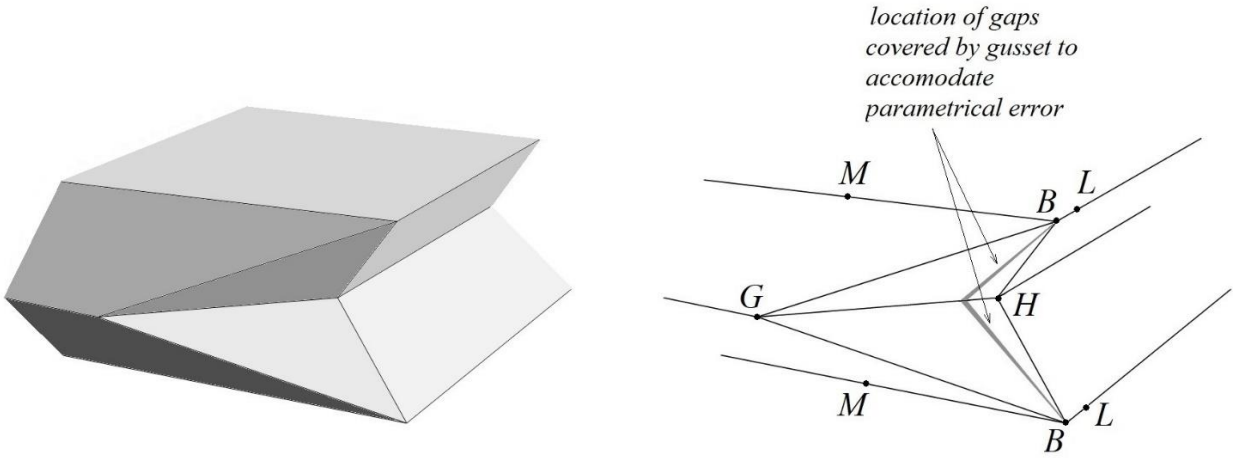


Figure 7

The perspective sketch at left shows how the two-facet alternate-folding corner would appear when fully inflated (i.e. with an inter-rib dihedral angle of 90°.) The sketch is shaded as if illuminated from the upper right, with shadows omitted for clarity.

The right side of Figure 7 shows the outline of the perspective sketch, with the key points labeled as in Figure 6. (Duplicate-labeled points correspond to the same point in the collapsed plan view of Figure 6) Also shown are the locations of the gaps in the rib fragments to accommodate the parametrical error, as they would be positioned if the banausic algorithm of section 8 were applied to the rib fragments *BHG*. In fact, the width of these gaps in Figure 7 is slightly exaggerated for visibility beyond what would be strictly necessary to accommodate the variation of the *GH* length over the range of permissible inflation heights of the reservoir corner.

Figure 8 presents elevation views of the inward- and outward-folding rib pairs of the two-facet alternate-folding corner.

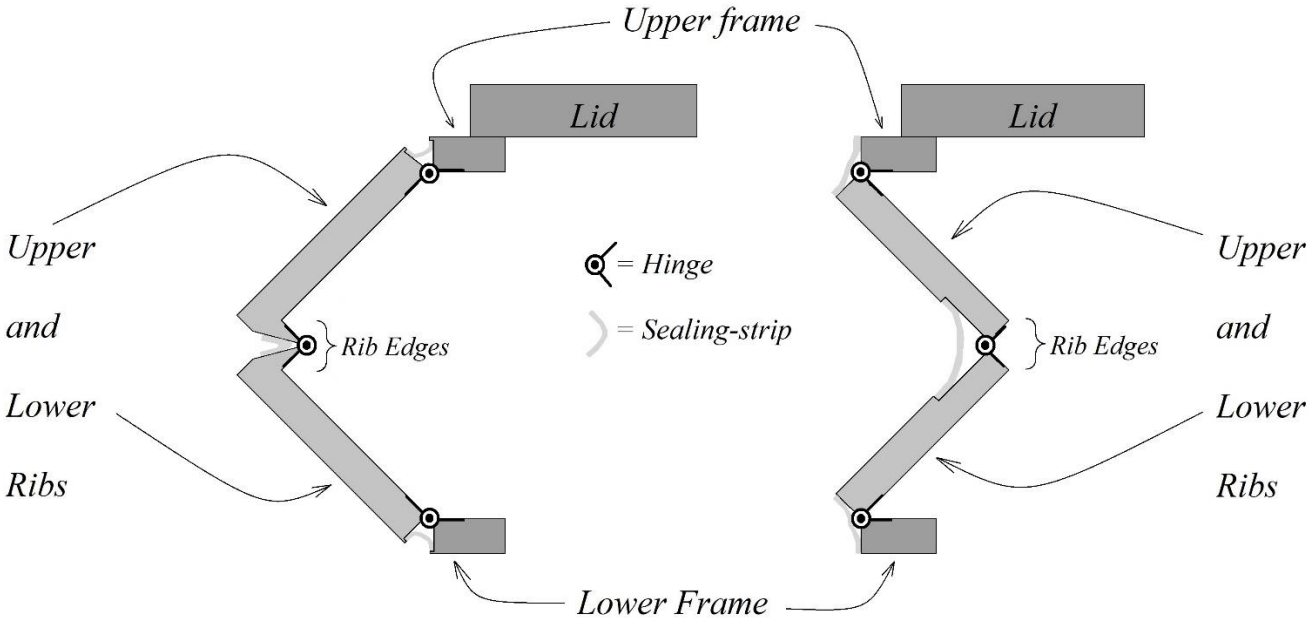


Figure 8

There are a few noteworthy features of the arrangement. Because some surfaces must fold flush against one another, the leather or rubber-cloth sealing strips along the joints may in some cases need to be inset into those surfaces. Because the inward-folding rib fragment pair fits between and separates the distal ends of

the outward-folding ribs, there is no need for a support nub as in Figure 5, but the outward-folding ribs must be crafted so that the rib fragments can fit between them, thus the odd cross-section of the outward-folding ribs.

If it be decided that some hinges affixing the outward-folding ribs to the upper frame should lie along the segment *BE* and within the area of *BHG*, then semicircular grooves or fossæ should be machined into the rib fragments to accommodate the hinge radii.

In real-world reservoirs, where the ribs and rib fragments are not of zero thickness, the air pressure acting on the rib edges will contribute to the distribution of forces within the reservoir at various inflation heights. Fortunately, the vector sum of the air pressure acting on edges and faces of equally-thick inward and outward rib pairs is equal to the vector of the air pressure acting on the face of comparably-sized zero-thickness rib pairs of slightly greater length and differing angle. The effect of non-zero rib thickness will be addressed in more detail in section 14 below.

5. Parametrical Error: Accounting for Variation in the Length of *GH* with Inflation Height

All the diagrams so far have shown the two-facet outward corner as seen in plan view from above in its full deflated and flat state.

Figure 9 shows a similar view of an alternate-folding corner to Figure 7, in which it is shown in its fully-inflated state.

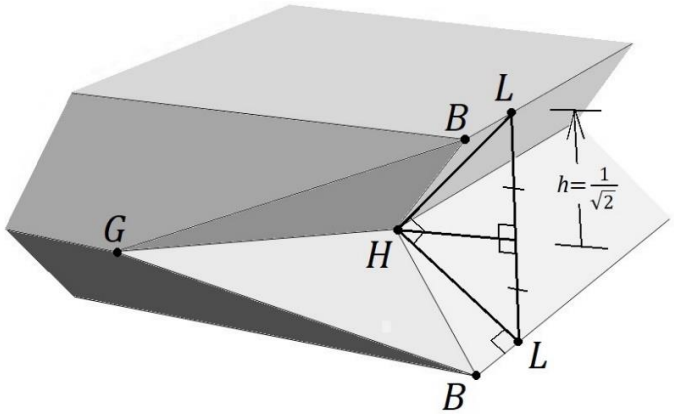


Figure 9

(machine-readable description of Figure 9: An alternate-folding two-facet corner is shown in perspective in its fully-inflated state, with labels as in plan view Figure 7. A right triangle whose corners are the upper L point L, lower point L, and point H, with the angle at H being the 90° angle. an altitude from point H is drawn to the opposite side, dividing the opposite side into 2 equal segments. The diagram is labeled so that the upper of these two equal segments is shown to have a height h of 1/√2)

Figure 9 illustrates that, at the end of its expansion over its permissible range of inflation, the height of the lid (containing the upper counterparts of points *B* and *L*) reaches a height *h* of $\frac{1}{\sqrt{2}}$ rib widths above the central plane that includes the rib fragment joint *GH*. Over the course of its inflation, from an inter-rib dihedral angle of 0° (flat) to 90° (maximum), point *G* along the outward-folding rib pair moves inward, while point *H* along the inward-folding rib paid moves outward.

Accordingly, as alluded to in regards to Figure 7, the length of *GH* changes over the course of the corner's inflation from an inter-rib dihedral angle of 0° to 90°, that is to say length *GH* is a function of h whose domain is $[0, \frac{1}{\sqrt{2}}]$. The *GH* line segment, itself, always remains in the $h = 0$ plane.

We define the *parametrical error* (notated ω) of a two-facet corner as the difference (measured, as always, in rib widths) between the longest length of *GH* and the shortest length of *GH* over its permissible inflation range $0 \le h \le \frac{1}{\sqrt{2}}$, from an inter-rib dihedral angle of 0° (flat) to 90° (maximum) when $d = a - \frac{1}{\sqrt{2}}$.

The convention of this paper is to notate parametrical error as plain ω, once the *a* value is chosen for a given reservoir design. By contrast, we define a *parametrical error function*, notated as $\omega(a)$, that maps from *a* values to parametrical error values when $d = a - \frac{1}{\sqrt{2}}$.

Using the value $d = a - \frac{1}{\sqrt{2}}$ to design a two-facet corner has three benefits:

(a) Of all the possible *d* values, $a - \frac{1}{\sqrt{2}}$ gives the smallest possible parametrical error for <u>any</u> *a* value chosen to design a two-facet corner, and regardless of whether outward-folding or alternate-folding.

(b) A desirable characteristic of a reservoir would be for the *GH* length to be at its maximum when fully deflated and also at its maximum at full reservoir inflation, with its minimum occurring intermediately between fully deflated and fully inflated. This gives builders or restorers a choice of two convenient configurations to apply the gusset to cover the gap necessitated by parametrical error. Of all the possible *d* values, only $a - \frac{1}{\sqrt{2}}$ gives this characteristic.

(c) For a chosen *a* value, when $d = a - \frac{1}{\sqrt{2}}$, we can calculate ω exactly as

$$\sqrt{(a-1)^2 + \left(a - \frac{1}{\sqrt{2}}\right)^2} - \sqrt{2}\left|a - \frac{2+\sqrt{2}}{4}\right|$$

We rigorously prove (b) as Corollary 3 in section 16 and (c) as Theorem 4 in section 17. Section 18 provides a sketch for a proof of (a) for $a > 1$.

For ease of convenience, the table below lists approximated parametrical errors for various values of *a*, taking, as always, the rib width as 1 (the rib width defines 1 unit.) (A more extensive table is provided in [2].)

a	***ω***	***a***	***ω***	***a***	***ω***
a_c	0.012609705	3.8	0.005143717	6.4	0.002733714
2.1	0.01212492	4	0.004817129	6.6	0.002638601
2.2	0.011229897	4.2	0.004529518	6.8	0.002549883
2.3	0.010457612	4.4	0.004274302	7	0.002466936
2.4	0.009784492	4.6	0.004046302	7.2	0.002389215
2.5	0.009192625	4.8	0.003841386	7.4	0.002316241
2.6	0.00866816	5	0.003656219	7.6	0.002247592
2.7	0.00820022	5.2	0.003488077	7.8	0.002182894
2.8	0.007780148	5.4	0.003334717	8	0.002121817
2.9	0.007400964	5.6	0.003194271	10	0.001657919
3	0.007056982	5.8	0.003065174	12	0.001360468
3.2	0.006456701	6	0.002946105	14	0.001153511
3.4	0.005950458	6.2	0.00283594	16	0.001001204
3.6	0.005517777				

Note that as the *a* value increases, the parametrical error decreases, with the parametrical error approaching zero as *a* approaches infinity. (We prove this behavior rigorously as Corollary 5 and Corollary 6 in Section 17.) This suggests, all else being equal, that the largest feasible *a* value should be used in designing a voided reservoir.

Referring to Figure 9, we define h_K (K: from German *kürzesten*, i.e. shortest[3]) as the *h* value which gives the minimum length of *GH* over the range $[0, \frac{1}{\sqrt{2}}]$ of inflation heights *h.* Lemma 11 and Corollary 4 in section 16 show that $h_K = \frac{1}{2} \cdot \sqrt{\frac{5}{2} - \sqrt{2}}$ when $d = a - \frac{1}{\sqrt{2}}$; h_K corresponds to an inter-rib dihedral angle of $2 \cdot \arcsin(h_K)$, approximately 62.799423963°, regardless of the corner's chosen *a* value.

The variation in *GH* length with *h* value when $d = a - \frac{1}{\sqrt{2}}$ is illustrated in more detail in Figure 25 in section 16.

6. A Simple Trimming Algorithm When ∠*GHB* is Acute

In [2], parametrical error was accommodated by trimming the rib fragments along the *BH* and *BG* edges, anticipating that the cloth hinge and leather or rubber-cloth gusset along those edges would serve to keep the rib fragments joined to the rest of the reservoir. However, as the reservoir height increases and the ribs and rib fragments need to be made of increasingly large, thick, and heavy materials, this arrangement becomes increasingly untenable, and one or more strong load-bearing metal hinges are preferable to join the ribs to the rib fragments along *BH* and *BG*.

In the adaptation of the two-facet outward corner presented here, we accommodate the parametrical error by dividing the triangular pieces (i.e. △*GBH*) into two *sub-fragments* with a central gusset-covered gap.

An intuitive way to do this would be to locate the altitude *BP* of △*GBH* from point *B,* and then remove triangular pieces *BMP* and *BNP* of base length $\frac{\omega}{2}$ from each side to form the two sub-fragments. A flexible gusset could then be placed to cover this gap. This procedure is shown in Figure 10, where the gap is greatly exaggerated for clarity.

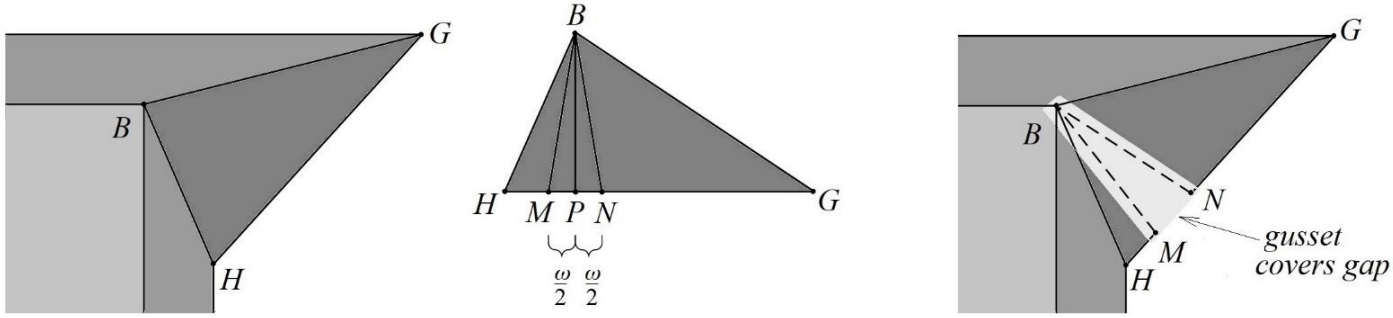


Figure 10

Lemma 3. A simple method of constructing an altitude of a triangle where the base angles ∠*GHB* and ∠*HGB* are both acute that confines the construction to the interior and edges of the triangle, is as follows, illustrated in Figure 11:

[3] *der h-Wert der kürzesten Länge* = the *h* value of the shortest length

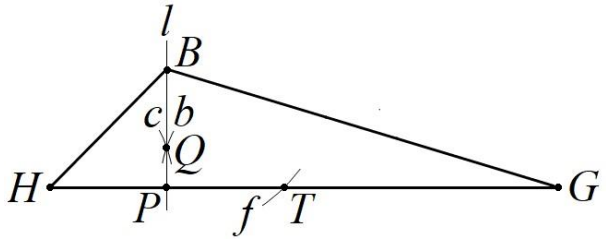


Figure 11

(machine-readable description of Figure 11: △ BHG, with ∠s at H and G both acute. Line l is drawn perpendicular to HG through point B. line l intersects segment HG at point P. Circular arc f centered on B has radius equal to BH. Arc f intersects segment PG at point T. Circular arc c centered on H has radius a bit larger than length of HP. Circular arc b centered on T has radius a bit larger than length of HP. Intersection of arc c and arc b is labeled point Q.)

1. Set your compass equal to length *BH* and draw arc *f* to locate point *T*
2. Set your compass a bit wider than length *BH*; center your compass on point *H* and draw arc *c*
3. At the same compass setting, center your compass on point *T* and draw arc *b*
4. Locate the intersection of arcs *c* and *b* and label that as point *Q*
5. Draw line *l* through points *B* and *Q*, extending line *l* to intersect *HG*
6. Mark the point of intersection of line *l* and segment *HG* as point *P*; *BP* is the altitude of △*HBG*

Proof of correctness. By construction, △*BHQ* is congruent (by SSS) to △*BTQ* . Hence ∠*QBH*=∠*QBT* . From this we conclude △*BHP* is congruent (by SAS) to △*BTP*, and ∠*BHP*=∠*BTP* , implying that ∠*BPH* and ∠*BPT* are supplementary right angles. ■

Note that other points besides the base of the altitude could be selected as *P*; indeed this is necessary when using *a* values that produce a rib fragment where ∠*GHB* is obtuse, right, or only slightly acute. In general, the further left *P* be chosen, the narrower the maximum gusset width, but the more chevron-shared the gusset tends to be.

While this simple method works quite well, it does not provide the absolute minimal gap between the sub-fragments, and thus requires a larger unsupported flexible gusset area than is actually needed. An even more precise algorithm when ∠*GHB* is acute is given in section 7.

Even so, the approach shown in Figure 10 (depicting an $a = 4$ corner) is really very good! For example, for a rib width of 915mm (about 3 feet), the gap would be only 4.41 mm wide at its widest at *MN*.

7. An Exact Banausic Trimming Algorithm When ∠*GHB* is Acute

Based on the trigonometry of the situation, one could derive the details of the ideal rib fragment triangle at the rib pair dihedral angle where *GH* length attains its minimum length. But, for *a* values greater than about 2.548 (above which the altitude *BP* lies inside △*GHB*) the following banausic algorithm works well to divide the rib fragments (△*GHB*) into sub-fragments with a minimal-width gap. As above, the widths of the slivers to be trimmed away are greatly exaggerated for clarity.

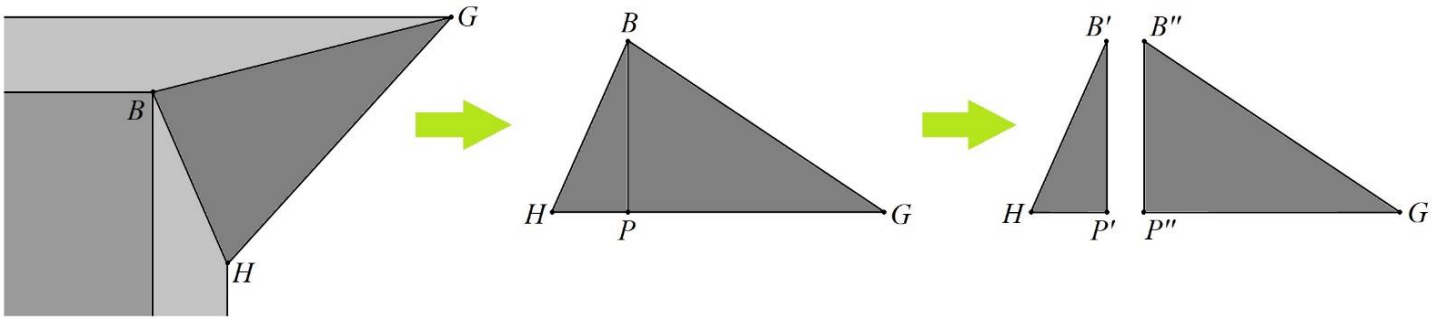


Figure 12

Figure 12 shows how the process begins. Starting with the rib fragments (i.e. triangles *GHB* cut out of wood or board) we first locate the altitude *BP* of each rib fragment. We cut this fragment along the altitude to create the sub-fragments *HB'P'* and *HB''P''*.

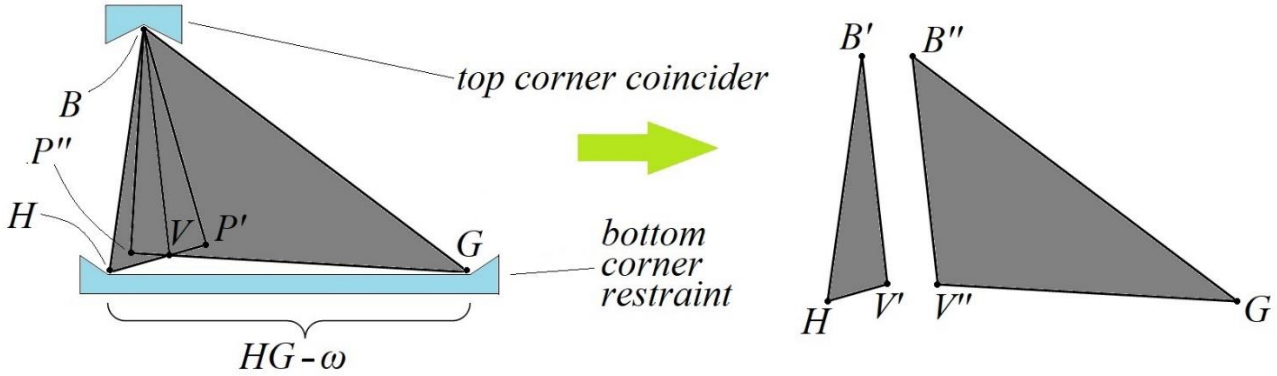


Figure 13

Figure 13 shows the process continuing. We should now manufacture a top corner coincider jig, which will hold the *B'* and *B''* corners coincident. We also manufacture a bottom corner restraint jig, which has a tray-like profile which will hold corners *H* and *G* at a distance from one another of $HG - \omega$.

With these appliances constructed, we can wedge the two sub-fragments among them as shown in the left diagram of Figure 13, supporting the upper fragment so it can't flop down. We locate point *V*, and mark its location on the edge of each sub-fragment (as *V'* on *B'P'H* and *V''* on *B''P''G*). We then cut away triangular sliver *B'P'V'* from sub-fragment *B'P'H* and cut away triangular sliver *B''P''V''* from sub-fragment *B''P''G*. In so doing, we produce trimmed sub-fragments *B'V'H* and *B''V''G*.

Finally, as shown in Figure 14, after we assemble the corner and connect the sub-fragments *BHV'* and *BGV''* to the ribs along *BH* and *BG*, we can cover the gap with a suitable gusset, knowing that the gusset will compress nicely over the whole range of 0° to 90° dihedral angle between each rib pair.

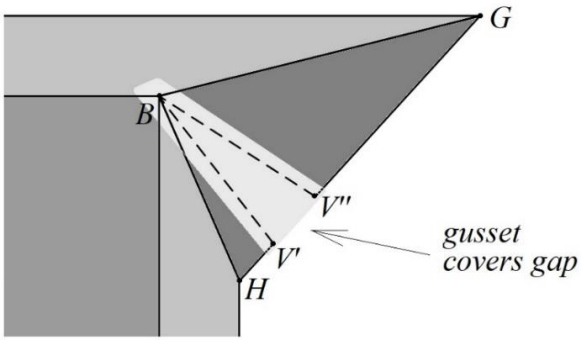


Figure 14

8. An Exact Banausic Trimming Algorithm When ∠*GHB* is Obtuse

As noted above, the minimum useable *a* value is a_c (2.05174376435691…) which gives a rib fragment whose altitude is equal to the rib width, allows commensurability between the rib fragments and the ribs, and is the most material-efficient corner geometry. However, because it has an obtuse angle at *H* we cannot simply divide the rib fragment into sub-fragments along its altitude and them trim the fragments as described in sections 5 and 6, because the rib fragment's altitude lies outside the rib fragment itself.

Figure 15 through Figure 17 illustrate a banausic algorithm for dividing a rib fragment based on an *a* value of a_c into two sub-fragments to provide a gusseted gap to accommodate the parametrical error inherent from such a large *a* value. These figures show the gap to exactly to scale, rather than exaggerated for clarity as in Figure 14. In fact, because $\angle GHB$ is obtuse, the gap created must actually be slightly larger than the parametrical error would suggest. This process is applicable to any *a* value (i.e. less than about 2.5480031964403…) where $\angle GHB$ is obtuse.

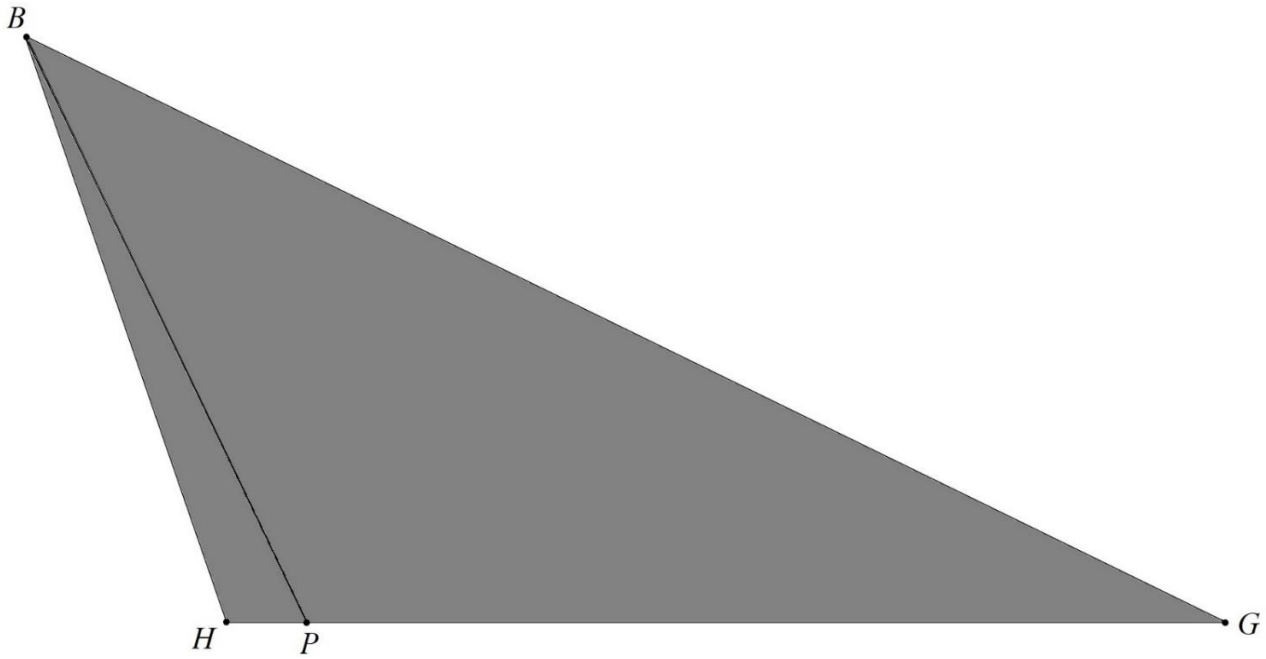


Figure 15

Referring to Figure 15: to begin the division, a point *P* should be chosen to mark the location of the bottom corner of the right-side sub-fragment. Importantly, the distance *HP* should be as small as possible while still providing enough space for machining recesses for the hinges along this joint with the main rib section and providing enough attachment surface area for the gusset that is to cover the gap. In the example illustrated in Figure 15 the distance *HP* was chosen to be approx. $6.39 \cdot \omega$, where ω is the parametrical error corresponding to a_c (about 0.0126097049774241.)

The rib fragment *BHG* is then cut along the *BP* line to form sub-fragment pieces *BHP′* and *BGP″* .

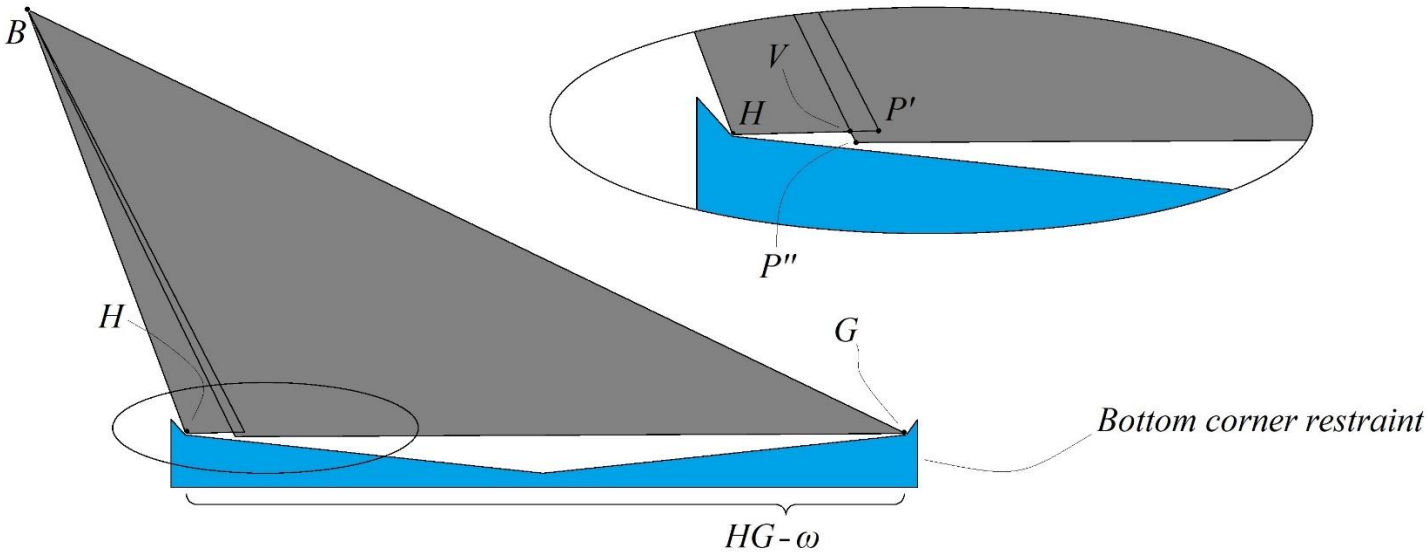


Figure 16

Figure 16 shows the process for trimming the left-side sub-fragment. Similar to the process shown above in Figure 13, the two pieces are wedged into a bottom corner restraint and a top corner coincider (not shown). Note that the bottom corner restraint must have a 'valley' to accommodate the fact that, for an obtuse triangular fragment, the right-side sub-fragment will protrude downward when the two sub-fragment pieces are wedged between the bottom corner restraint and top corner coincider. As in Figure 13, the length of the valley of the bottom corner restraint will be $HG - \omega$.

The inset oval of Figure 16 shows an enlarged view of the overlap region between the two sub-fragment pieces. To assure proper fit, we mark the location of the intersection point *V* on the edge of the left-side sub-fragment *BHP′* and then remove the narrow triangular sliver *VP′B* off of *BHP′* . In so doing, in this example *HV* ends up with a length of approx. $5.14 \cdot \omega$, giving a *VP′* distance being approx. $1.25 \cdot \omega$.

It is also advisable to round over the corner slightly around point *P″* to minimize poking of *P″* against the gusset material.

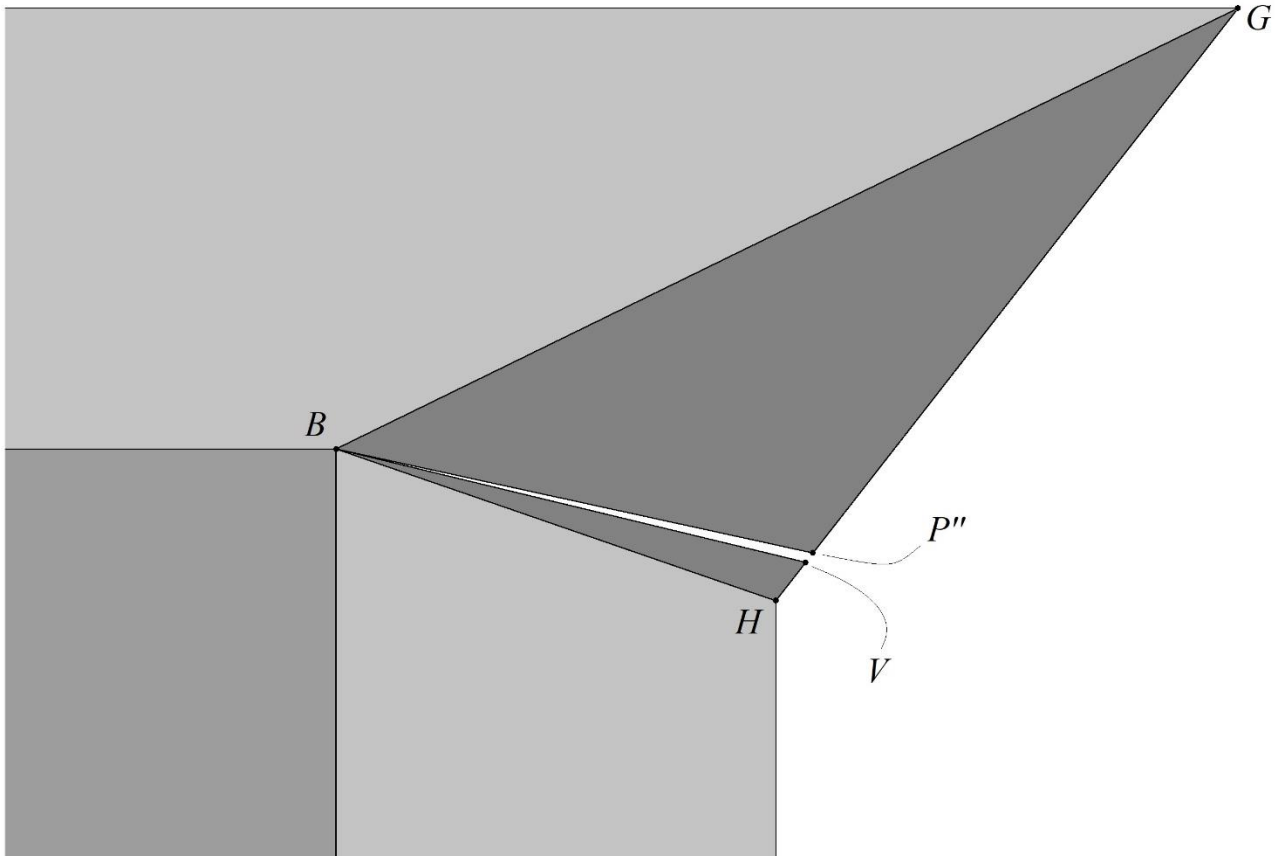


Figure 17

Figure 17 shows how the resulting sub-fragment pieces *BHV* and *BP″G* are assembled into a completed corner geometry. Even with its *a* value of a_c , this corner's gap that must be covered by a flexible gusset (not shown) is fairly narrow -- far narrower than the usual gusset-covered gap in a conventional reservoir corner design!

9. Size Calculations for a Standard Voided Reservoir

Figure 18 show a reference diagram for designing a standard voided reservoir of the type introduced in [2], using two-facet outward-folding corners. To maximize capacity, the main reservoir chamber is typically (although not necessarily) square, as indicated in Figure 18, while the void space is typically designed to be as narrow as possible consistent with the aerodynamics of easy air flow through the slot in the trunk-band floor under the void space. The relationships of the main reservoir chamber and void space dimensions are indicated in the figure.

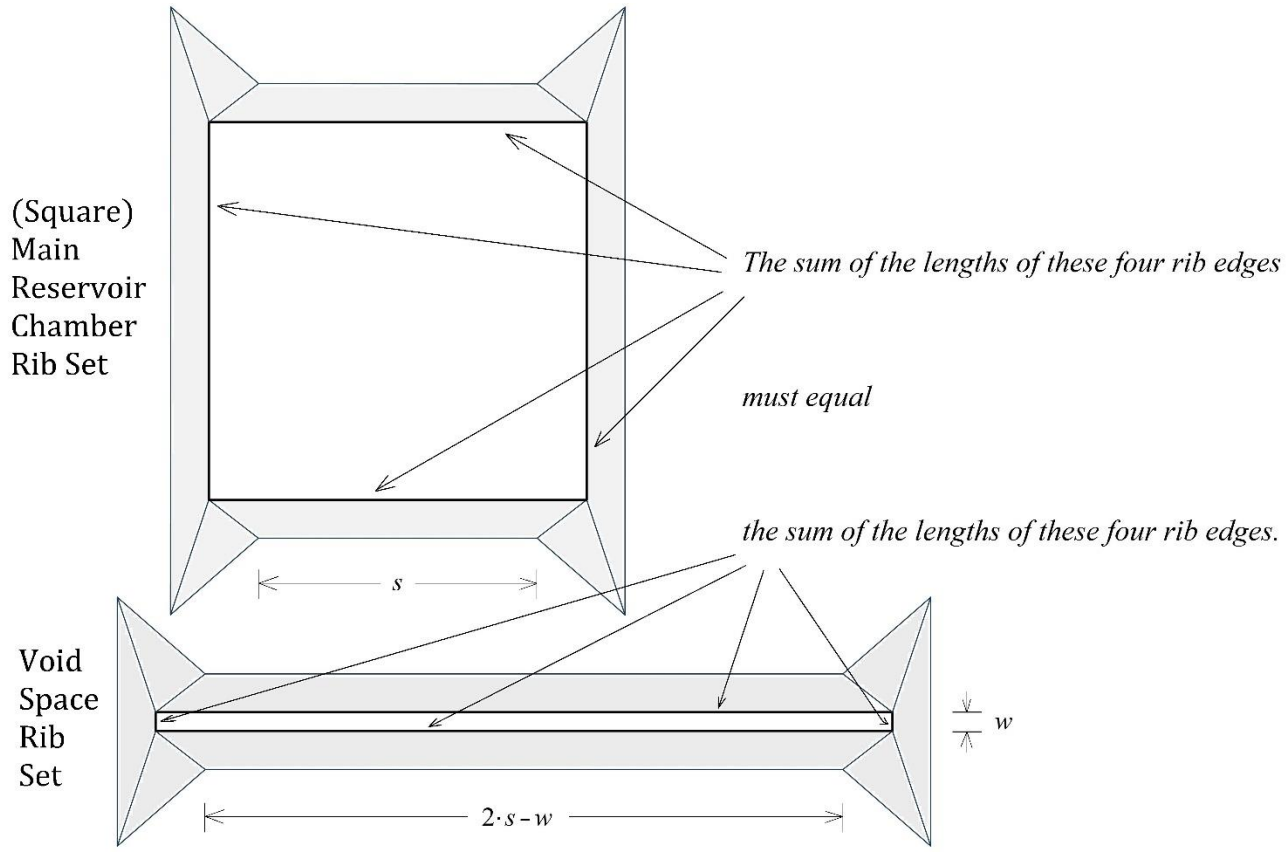


Figure 18

The capacity of the voided reservoir is not simply the main chamber reservoir lid area multiplied by the maximum inflation height. Rather, it is this volume minus the void space lid area multiplied by the maximum inflation height. We must account for this in designing a voided reservoir to have given capacity.

The length *s* of each side of the main reservoir chamber lid in Figure 18, taking into account the dimensions of the void space, must satisfy the equation:

$$s^2 = A + w(2s - w)$$

Solving[4] gives the delightfully simple result that

(1) $$s = \sqrt{A} + w$$

Where *A* is the main reservoir chamber's lid area, and *w* is the width of the void space.

For example, if a reservoir is to have a capacity of 320 cubic meters at full inflation, and the rib width is to be 1.2 meter, then the reservoir's height at full inflation will be about 1.7 meters, meaning its main reservoir chamber's lid area must be about 189 square meters. The void space is to have a width of 0.5 meter.

Using formula (1), this would indicate each rectangular lateral rib area should be of length 14.24773 meters.

[4] Algebra credit: Wolfram Alpha LLC. 2026. Wolfram|Alpha. http://www.wolframalpha.com (access August 11, 2026).

10. A Thought Experiment: Designing a Fantastically Huge Voided Reservoir

To get a sense of how effective the two-facet corner design can be in terms of minimizing the width of the gaps at the corners to be covered by flexible gussets, so that nearly the entire rib-band surface area can be constructed from rigid material, consider the following scenario:

Suppose it is desired to build a very large symphonic pipe organ to play day-long concerts, such as where a transcription of one or more videogames' full music score is performed by a trio of organists from three consoles. A typical concert would have about 5½ hours of music, to be presented in seven sections of approximately 47 minutes each, separated by six 15-minute intermissions.[5]

And imagine (by way of maximizing the design capacity remit) that a voided reservoir is to be constructed with enough capacity that it could be filled fully in advance of the concert, and then have additional air added during the intermissions, with the blower(s) being turned off and their inlets closed during the playing portions of the concert. Finally, assume the blower(s) have a combined capacity of 1100 cubic meters (~38850 cubic feet) per minute with the organ consuming on average 425 cubic meters of wind per minute of play over the course of a typical concert.

This works out, with an additional margin of about 5% to compensate for leaks[6], to a reservoir having a useful capacity of about 43,000 cubic meters of air!

Such an enormous reservoir, even without periodic additions of more air during the course of a concert, is still large enough to provide 96 minutes of continuous music.

The space for the reservoir, located under the vast concert hall, is large enough that a reservoir with a total height of 12.25 meters could be fit. Accordingly, each main reservoir chamber and each void space could have an internal maximum inflation height of about 5.125 meters, for a rib width of 3.624 meters.

The designers, keenly aware of the practical problems of building and weighting vast main reservoir chamber lids, may decide to build the reservoir by dividing its total capacity into four square main reservoir chambers. For simplicity, the entire organ could be on a single, relatively low wind pressure.

[5] Food and drink, however, would be available continuously during the concert from catering stations in the auditorium. 3-D videos of relevant episodes from the videogames may also be projected on a large screen, viewable with 3-D glasses.
[6] Hopefully relatively little leakage, thanks to diligent reservoir construction!

The plan view of such a compound voided reservoir is shown in Figure 19.

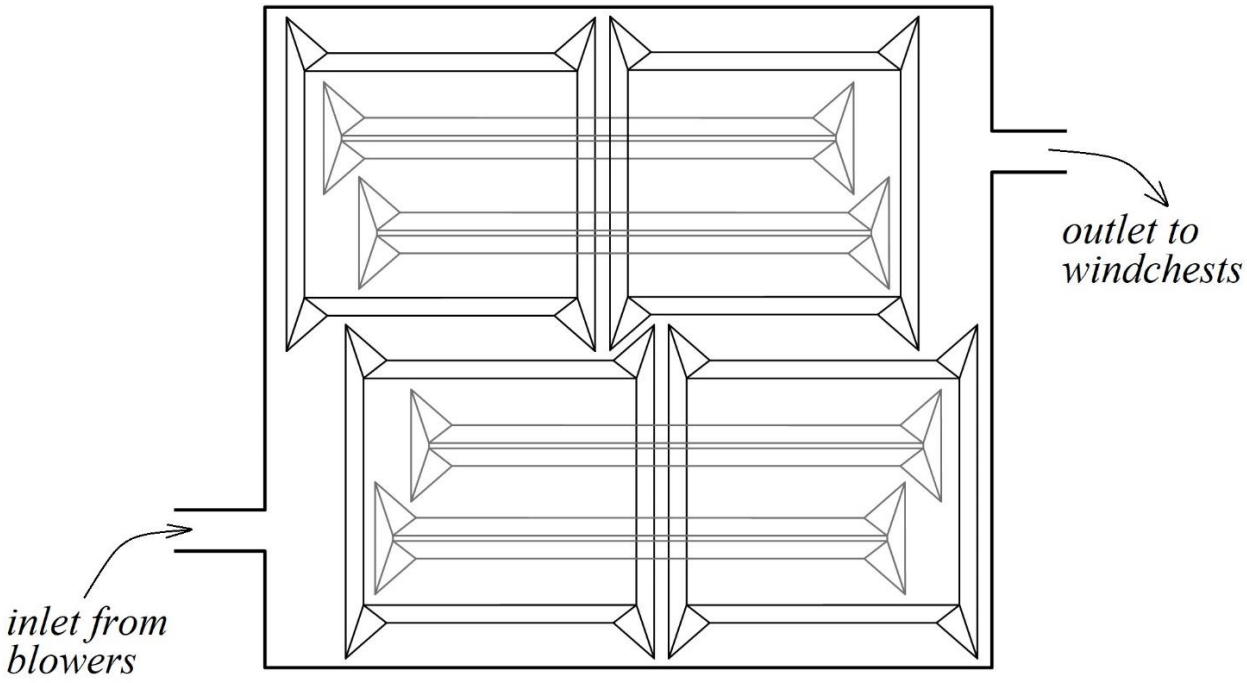


Figure 19

The main trunk band, which provides inlets for the blower(s) and outlets to the various windchests, is demarcated by the thick solid line. The four main reservoir chambers are demarcated by the thin black lines. The four void spaces, demarcated in thin gray lines, all fit within the plentiful volume of the trunk band.

With the main reservoir chambers having an internal height of about 5.125 meters, the total active lid area should be 8390.25 square meters. Thus, each of the four chambers will have an active lid area of 2097.56098 square meters. If we further assume each void space will be a quarter as wide as the rib width, that means each void space will be 0.906 meter wide. The main reservoir lid will have a thickness of 1.5 meters, with the ribs and void space lid having a thickness of 14cm.

Putting these dimensions into formula (1) the designers learn that each of the four square main reservoir chambers would be 46.705 meters along each side.

For this project, the designers may decide to design the reservoir with an *a* value of 3, this giving a fairly narrow gap to be covered with the sub-fragments' gap's gussets and the chance to use the algorithm presented in Figure 12 through Figure 14, while still using comparably little additional rib-band construction material beyond what would be needed if an *a* value of a_c would have been used.

An *a* value of 3 gives parametrical error ω of ~0.007056982. This works out to a gap at each corner of approx. 25.57mm (a bit over one inch) at its widest (i.e. distance from *V'* to *V''* in Figure 14). This may be considered quite acceptable considering the enormous overall dimensions of the reservoir.

11. A Single-Level Voided Reservoir Enabled by Two-Facet Alternate-Folding Corners

Figure 20 introduces a novel variation of the voided reservoir, facilitated by the use of two-facet alternate-folding corners, and shows how it is conceptually assembled. This novel reservoir design is shown in plan view in its fully-collapsed state. In this design, the void space is not located on the floor of the trunk-band below but instead is integrated as a 'hole' in the reservoir's main chamber itself.

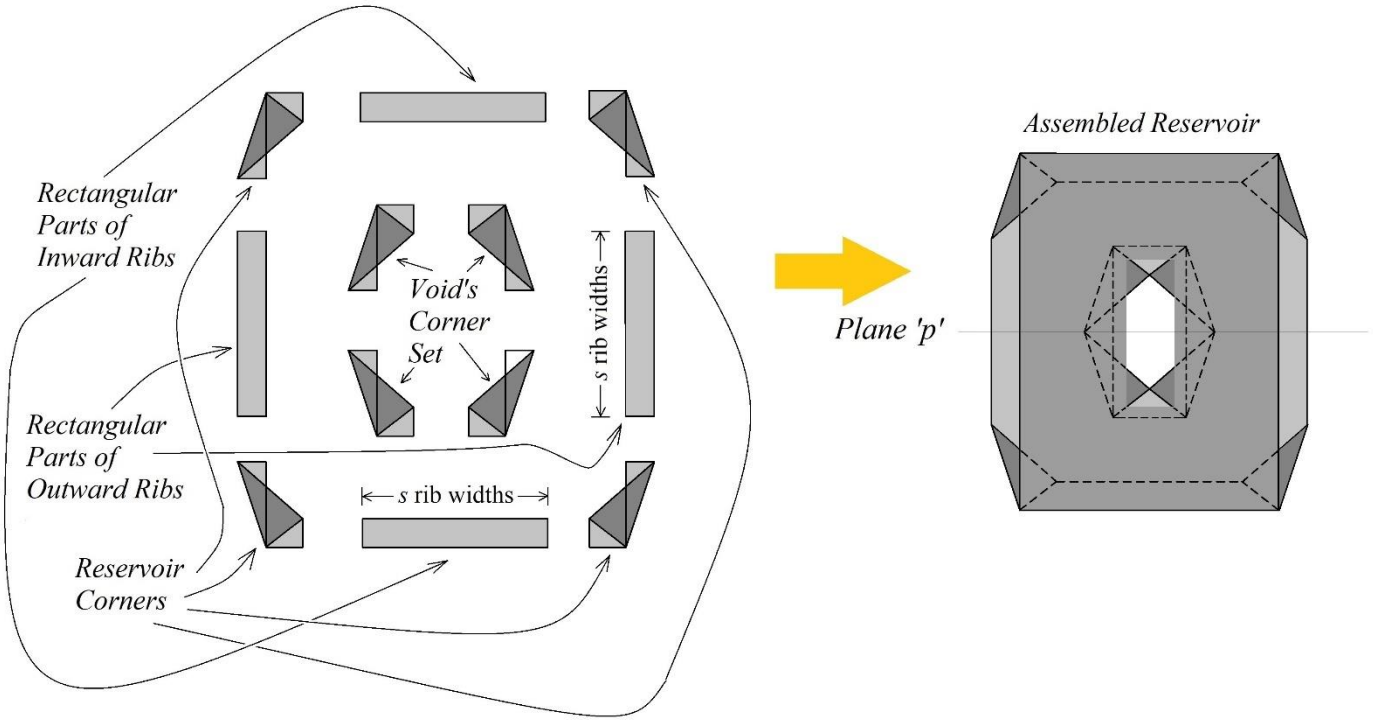


Figure 20

It should be noted that this design does not require the *a* value of the corners to be a_c; any reasonable value will work. (The corners of Figure 20 are constructed with an *a* value of 3) But the larger the *a* value, the larger the central void space must be, and the less capacity-efficient the reservoir will be with respect to its footprint. The author believes these issues are offset by the material cost savings, simpler construction, and greater volume of pressurized area for a given total reservoir volume, compared to the original two-level voided reservoir concept.

The left portion of Figure 20 shows that, conceptually, the outer rib-band of the reservoir is assembled from four alternate-folding corners, two opposite pairs of outward-folding rectangular rib area, and two opposite pairs of inward-folding rectangular rib area. The rectangular rib areas are all identical and of length *s* rib widths in the design formula below. Of course, in practice the distal triangular portions of the lateral ribs would probably not be machined separately -- this diagram represents a conceptual, rather than practical, assembly diagram. The key is that the rectangular rib areas must all be equal, so the air pressure effects on the outward-folding rectangular rib areas exactly cancel the air pressure effect on the inward-folding rectangular rib areas.

Likewise, the air pressure effect on the four alternate-folding corners that compose the rib-band of the void space exactly cancels the air pressure effect on the four alternate-folding corners along the outside of the reservoir.

The right portion of Figure 20 shows how these conceptually separate components are assembled into a completed reservoir.

A reservoir constructed in this manner, to within the tiny variation introduced by the varying-size gusset gaps, causes the combined internal air pressure forces on all the surfaces and edges of all the rib-band components (both outer and void space) to have a net zero force on the reservoir lid, regardless of inflation height.

Figure 21 shows a schematic elevation view of such a reservoir as it would appear sliced at the plane 'p' in the right drawing of Figure 20.

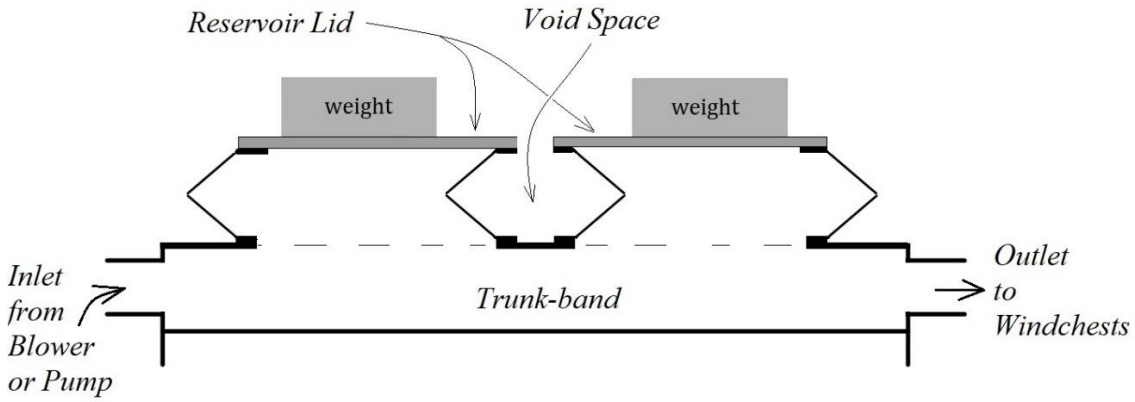


Figure 21

Note that although the void space appears to float in this diagram, it would be attached to the reservoir trunk-band exactly at the level that the outer rib set would be. The solid reservoir lid, being common to the outer rib set and void rib set, fulfills the same functional role as the rigid linkage between the main reservoir chamber and the void space in the original voided reservoir design.

This elevation view makes it more apparent why the outward-folding ribs would be attached to the frames (cf. Figure 8) in this design, rather than directly to the reservoir lid: parts of the outward-folding triangular rib-pieces of the void space would otherwise have to physically impossibly coincide with the reservoir lid!

Designing this type of reservoir is made easier by expressing all the dimensions in terms of rib widths.

The goal is to find a length *s* of each rectangular rib area in Figure 20 so that the outside dimensions of the lid will be large enough to have the desired active lid area, even after accounting for the void space top area subtracted from the total lid area. Such a value of *s* (measured in rib widths) satisfies the equation:

$$(s + 2a)\big(s + 2(d - 1)\big) = A + 2a \cdot 2(d - 1)$$

Where *a* is the *a* value, *d* is the *d* value (typically $a - \frac{1}{\sqrt{2}}$) , *A* is the active lid area, and *f* is the horizontal width of the frame, all measured in rib widths or square rib widths.

Solving[7] gives the explicit formula:

(2) $$s = \sqrt{a^2 + 2a(d - 1) + A + (d - 1)^2} - a - d + 1$$

For example, if a reservoir is to have an *a* value of 3 and a *d* value of $3 - \frac{1}{\sqrt{2}}$, a capacity of 320 cubic meters at full inflation, and the rib width is to be 1.2 meter, then the reservoir's height at full inflation will be about 1.7 meters, meaning its active lid area must be about 189 square meters. With a rib width of 1.2 meters, that is an active lid area of 131.25 square rib widths. The frames will have a width of 0.2 meters, or 0.16666… rib widths.

[7] Calculation and algebra credit: Wolfram Alpha LLC. 2026. Wolfram|Alpha. http://www.wolframalpha.com/input/?i=2%2B2 (access August 17, 2026).

Using formula (2), this would indicate each rectangular lateral rib area should be of length 7.9414 rib widths, or 9.53 meters. Thus, we arrive at an actual reservoir lid of dimensions approx. 16.73 meters × 15.03 meters.

The 'hole' in the lid will have dimensions in rib widths of

$$2 \cdot (a - f) \times 2 \cdot (d - 1 - f)$$

12. Designing the Fantastically Huge Reservoir with Alternate-Folding Corners

We now revisit the thought experiment of section 10, where the remit was to design a reservoir with a capacity of about 43,000 cubic meters of air, but using the design introduced in section 11. Figure 22 shows a plan view for how such a reservoir could be configured.

Figure 22 and Figure 19 are roughly to the same scale.

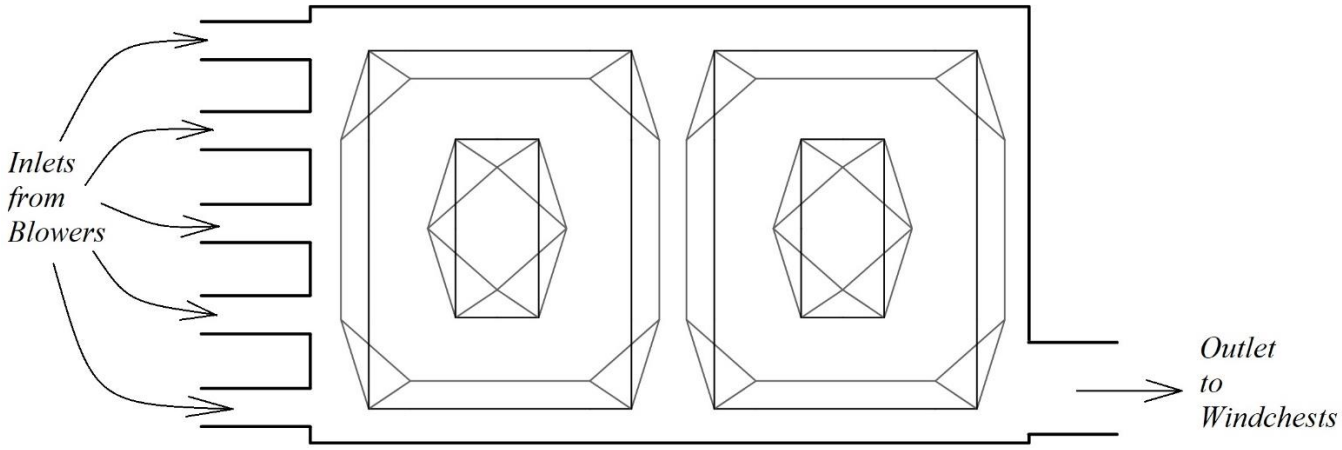


Figure 22

Beginning from the same total height constraint (12.25 meters), the designers decide that the reservoir will need a lid thickness of 2 meters for sufficient rigidity and the trunk band will be 2.23 meters high, accommodating the massive wind-lines from the blowers and to the windchests. That leaves 7.5 meters for the inflation height of the reservoir to maximum capacity, implying a rib width of 5.31 meters. The designers decide to divide the capacity among two reservoir chambers on the common trunk-band. The frames, as well as the rib material itself, will be 14cm thick.

For a reservoir height of 7.5 meters and 43,000 cubic meters capacity, this gives an active weighted surface area of 5734 square meters: 2867 square meters for each of the two reservoir chambers.

The designers decide to use the generous *a* value of 3.2 which gives a parametrical error ω of ~0.006456701. For a rib width of 5.31 meters, this works out to needing to cover 35mm[8] (or a bit less, if they use banausic algorithm of section 7) at the gap's widest point (i.e. gap *V'V''* in Figure 14) at each *BG* edge joint with leather gusset material. The designers believe this is acceptable and will not result in excessive pouching or wear of the gusset.

Turning their attention to the sizing of each of the two reservoir chambers: with a rib width of 5.31 meters, the active weighted surface area of each reservoir chamber will be 101.7 square rib widths. The frames' horizontal width (viz. Figure 8) is chosen as 0.65 meters, which is 0.1225 rib widths.

[8] Cf. a large USA Eisenhower dollar coin from the 1970's at 38.1mm

Working from the formula (2) introduced in section 11 above, the value of *s* is found to be 6.4302 rib widths, or about 34.1444 meters. Thus, we arrive at outside dimensions of the lids for each of the two reservoir chambers of 68.129 meters × 50 meters.[9]

13. Commensurable Two-Facet Alternate-Folding Corners

Section 3 introduced the notation a_c (c: commensurable) for the special *a* value approximately 2.0517…, noting that it allowed the rib fragment to be commensurable, in terms of effect of air pressure upon its faces, as a region of lateral rib of equal surface area.

Lemma 4. A two-facet corner constructed with an *a* value of a_c has a surface area of $2a_c$.

Proof. Regarding the trapezoid *DGHL* of Figure 6, we can observe its two base lengths are *a* and 1, while its height *d* is $a - \frac{1}{\sqrt{2}}$ as usual. Thus, its area is

$$\frac{1}{2}(a+1)\left(a-\frac{1}{\sqrt{2}}\right)$$

Setting this expression equal to *a* to solve for the case where the area of the trapezoid *DGHL* equals *a* itself, we find that a_c is the positive root.[10] Thus, the corner (i.e. the trapezoid *DGHL* consisting of the extensions $\triangle BDG$ and $\triangle BHL$ of the ribs and the rib fragment $\triangle BGH$) constructed with an *a* value of a_c has an area of $2a_c$, from the inclusion of the two portions (upper and lower) of the corner. ■

Corollary 1. An outward-folding corner constructed with an *a* value of a_c and *d* value of d_c is equivalent in terms of air pressure effect to a segment of outward-folding lateral rib face of width 1 and length a_c when it is constructed of zero-thickness material.

Proof. In an outward-folding corner, $\triangle BDG$ and $\triangle BHL$, and rib fragment $\triangle BGH$ are outward-folding surfaces. Since their total area is a_c, their altitudes are all 1 rib width, and their angles are all the same as the lateral ribs' dihedral angle, we find their total area is equivalent in terms of air pressure effect (neglecting thickness) as a segment of lateral rib area of length a_c and width of 1 rib width. ■

Lemma 5. In a two-facet corner, $\triangle MGB$ and quadrilateral *BGHL* are equal in area when the corner is constructed with an *a* value of a_c.

Proof. From Figure 6, we see that trapezoid *DGHL* consists of $\triangle DGB$ and quadrilateral *BGHL*. When the corner is constructed with an *a* value of a_c, the area of $\triangle DGB$ is $\frac{1}{2} \cdot a_c$. Since the area of the trapezoid *DGHL* is a_c, the area of quadrilateral *BGHL* must also be $\frac{1}{2} \cdot a_c$.

By ASA, $\triangle DGB$ is congruent to $\triangle MBG$, so $\triangle MGB$ also has area $\frac{1}{2} \cdot a_c$. ■

Theorem 1. An ideal zero-thickness two-facet alternate-folding corner with *a* value equal to a_c has a net zero effect on the pressure variation of a reservoir with inflation height, to within the tiny variation introduced by the varying-size gusset gap.

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[9] Calculation credit: Wolfram Alpha LLC. 2026. Wolfram|Alpha. http://www.wolframalpha.com (access August 11, 2026).

[10] The negative root produces the same corner geometry but reflected around *y=x*

Proof. Observe that $\triangle MGB$ is exactly the distal portion of the outward folding rib, and quadrilateral *BGHL* is exactly the distal portion of the inward-folding rib combined with the inward folding rib fragment. By Lemma 5, the areas of these outward and inward folding surfaces are equal. ■

So an ideal zero-thickness two-facet alternate-folding corner with *a* value equal to a_c has a net zero effect on the pressure variation of a reservoir with inflation height, to within the tiny variation introduced by the varying-size gusset gap. This result will be applied in section 14:

The pressure variation of a reservoir built around such two-facet alternate-folding corners would depend exclusively on the relative areas of the central rectangular regions of the lateral rib-pairs – assuming zero-thickness rib material. We apply this result in section 14 to a novel type of imperfect camera-folded reservoir with real-world non-zero rib material thickness.

Lemma 6. There is a pleasing relationship between the areas of the trapezoid *DGHL* and its constituent triangles:

	$\triangle BGH$		$\frac{2(2+\sqrt{2})}{8}$		
	$\triangle BDG$	$\frac{(2+\sqrt{2})}{8}$		+	$\frac{\left(\sqrt{6+12\sqrt{2}}\right)}{8}$
+	$\triangle BHL$	$\frac{-(2+\sqrt{2})}{8}$		+	$\frac{\left(\sqrt{6+12\sqrt{2}}\right)}{8}$
=	Trapezoid $DGHL$		$\frac{2(2+\sqrt{2})}{8}$	+	$\frac{2\left(\sqrt{6+12\sqrt{2}}\right)}{8}$

Proof. To find the area of rib fragment *BGH*, we can write the area of the trapezoid *DGHL* as the of the constituent triangles' *BGH*, *BDG*, and *BHL* areas:

$$\text{area} \triangle BGH \; + \; \frac{1}{2}a \; + \; \frac{1}{2}(d-1)$$

Being the sum of the constituent triangles *BGH*, *BDG*, and *BHL* areas. When we plug in the special values a_c for *a* and d_c for *d*, and set this sum equal to a_c as above, we can solve to find the area of rib fragment *BGH* to be

$$\frac{2+\sqrt{2}}{4}$$

It then is a matter of multiplying this area by $\frac{2}{2}$ and rearranging and restating the areas of trapezoid *DGHL* and $\triangle BDG$ in terms of a_c. The area of $\triangle BHL$ is then found by subtraction. ■

14. An Imperfect Camera-Folded Reservoir Enabled by Commensurable Two-Facet Alternate-Folding Corners

Section 13 hinted at the idea that by using two-facet alternate-folding corners with *a* value equal to a_c (which gives equal areas of inward- and outward-folding rib material) a reservoir could be constructed whose pressure variation depends only on the relative areas of the central rectangular regions of the lateral rib-pairs.

If the relative areas of these regions could be equal (i.e. a square reservoir) we would have built a camera-folded reservoir (cf. the diagram in [1]) anticipating an overall essentially zero net effect on the pressure variation inflation height. Figure 23 illustrates this concept.

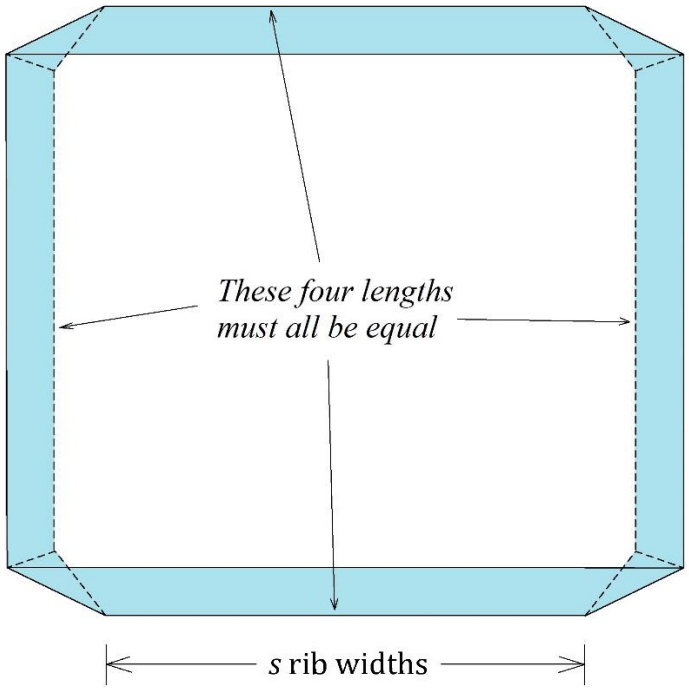


Figure 23

Note that this design is not a kind of voided reservoir, but a special type of single-rise weight-driven reservoir. The construction is much simpler: with no void space, the outward-folding ribs could be attached directly to lid, rather than needing an intermediate frame (cf. Figure 8)

Analogously to the design equation in section 11, The length *s* (the lengths, in terms of rib widths, labeled as "These four lengths must all be equal" in Figure 23) of each rib's short side in Figure 23 satisfies the equation:

$$(s + 2a_c)\big(s + 2(d_c - 1)\big) = A$$

Where *A* is the active lid area, measured in square rib widths. Solving[11] for *s* we obtain:

(3) $$s = \sqrt{{a_c}^2 - 2a_c(d_c - 1) + A + (d_c - 1)^2} - a_c - d_c + 1$$

It may seem that this is a perfect solution to the constant-pressure weight driven reservoir design question first posed in 2000. And it would be, if all the ribs and rib fragments had zero thickness. To understand the

[11] Algebra credit: Wolfram Alpha LLC. 2026. Wolfram|Alpha. http://www.wolframalpha.com (access August 9, 2026).

imperfection of this design, we can analyze and compare its mechanics to that of the original voided reservoir design of [2] and to that of the alternate-corner voided reservoir design of section 11 above.

Theorem 2: In the original voided reservoir design of [2] and the alternate-corner voided reservoir design of section 11 above, the pressure is constant over the entire inflation range to within the variation introduced by the varying length of the *GH* joint and the changing shape of the flexible gusset over the gap between the rib fragments comprising *BGH* and over other corner and edge gaps.

Proof. In any reservoir, internal air pressure forces act upon the ribs and rib fragments to wedge open (for a concave dihedral with respect to internal air pressure) or squeeze shut (for a convex dihedral with respect to internal air pressure) the reservoir. Thus, each dihedral surface pair may counteract or cooperate with the weight's force on the reservoir lid to set the reservoir's internal pressure.

Consider the original voided reservoir design described in [2]. For every unit of surface area participating in a given concave dihedral angle with respect to, and exposed to, internal air pressure there is a corresponding unit of surface area participating in an equal convex dihedral angle with respect to, and exposed to, internal air pressure. This is true whether the unit of surface area belongs to a face, an edge, a rib or a rib fragment.[12] As a result of this opposing correspondence, the reservoir lid feels no net effect from changing rib set angular position with inflation height.

This complete correspondence is true also for the alternate-corner voided reservoir design of section 11: The main difference being that with respect to the faces and edges of the rectangular portions of the ribs, for every unit of inward folding rib surface area exposed to internal air pressure, there is a corresponding unit of outward folding rib surface area among these portions (viz. Figure 8) that is exposed to internal air pressure. Meanwhile, the corner components set up a correspondence as in the original voided reservoir corners.[13] ■

Due to the use of the commensurable corner species, all the faces in the non-voided camera-folded reservoir design of the present section participate in equal dihedral angles. But it is ***not*** the case that every unit of surface area participating in a concave dihedral angle with respect to, and exposed to, internal air pressure has a corresponding unit of surface area participating in an equal convex dihedral angle with respect to, and exposed to, internal air pressure: these unmatched surface areas are exactly the *GH* joint edges (viz. Figure 7) forming a small outward-folding or concave dihedral at each corner from the point of view of internal air pressure.

Thus, the non-voided camera-folded reservoir design of the present section, constructed from non-zero thickness materials, does vary a bit in pressure over its inflation range.

In general, a variation of 1% in pressure may be acceptable for narrower-scale pipes such as narrow-scale flutes, principals, and strings, and full-length reeds, while a variation of 0.5% or less may be considered good, and essential for more pressure-sensitive pipe types such as fractional-resonator reeds and wide scale flutes (e.g., tibia clausa pipes). The author has not modeled the amount of pressure variation over the inflation range of this reservoir design due to the unbalanced air pressure forces on those rib edges at the corners. But, based on rough estimates, he believes that when the reservoir's length and width exceed 12 rib-widths and the permissible inter-rib dihedral range is limited to 10° to 90°, this reservoir design may achieve or even surpass such tight pressure variation requirements when the rib thickness is 10% or less of the rib width.

---

[12] The correspondence among edge areas occurs completely within the void space rib set.

[13] Although the description is in terms of discrete vector forces on faces and edges, the argument also applies when the air pressure in the reservoir is thought of as a continuous normal-vector field impinging in the upon the non-rectilinear profile of the outward-folding rib components.

Restricting the permissible inter-rib dihedral range to 30° to 90° allows even greater pressure stability.

Under such constraints, this reservoir design appears to perform at least as well as conventional single-rise inward-folding reservoirs compressed by coil springs.

Using formula (3) to work the same example as above: if a reservoir is to have a capacity of 320 cubic meters at full inflation, a rib width of 1.2 meter. Let's further assume that we'll constrain its range of inflation height to start from an inter-rib dihedral of 10°, rather than 0°. This gives a range of reservoir height $2 \cdot 1.2 \cdot sin(5^0)$ when empty to $2 \cdot 1.2 \cdot sin(45^0)$ when full, or about 1.48788 meters. Then the active lid area about 215.0711 square meters, or 149.354936 square rib widths. We find that each side s must be 9.94336 rib widths, or 11.932 meters. This gives an actual reservoir lid of dimensions approx. 12.759 meters x 16.856 meters. The gap at the corners to be covered by a gusset would be a bit under 19mm at the widest point along the *GH* joint (i.e. gap *VP''* in Figure 17) in accordance to the banausic trimming algorithm of section 8.

In summary, if one is willing to build a reservoir that is quite long and wide relative to its height, and reduce its capacity by not allowing it to deflate completely (i.e. defining 'empty' as already open to a 10° or even 30° inter-rib dihedral angle) pressure variation of under 1% over the permissible inflation range of a reservoir or regulator seems achievable with the non-voided design described in this section. But if one is willing to accept more construction complexity, in exchange for higher capacity for a given footprint and essentially zero pressure variation, and enjoy choice of *a* value, the alternate-corner voided reservoir design of section 11 may be a better choice. On the other hand, if the floor space available for the reservoir is very large, and the thickness of the ribs is very small compared to their width, but the available height is quite small, the non-voided design described in this section may be preferable. Designers may also consider stacking several reservoirs vertically of the non-voided design described in this section, to minimize the rib width while increasing total capacity.

15. Deriving an Exact Expression for a_c

Figure 24 Illustrates the situation when the distance from the point $B\ (0,0)$ to the line determined by points *G* and *H*, namely distance *BZ*, is exactly one rib width. *BZ* is the altitude of the rib fragment $\triangle BGH$.

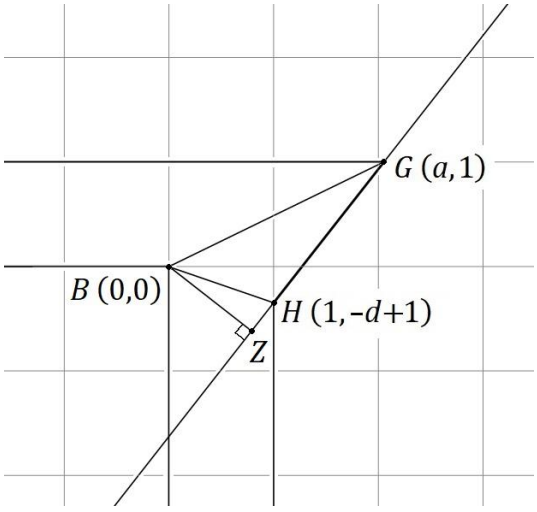


Figure 24

(machine-readable description of Figure 24: Upon a Cartesian coordinate system with an origin B (0,0): A horizontal ray with its rightmost point G at (2.0517,1). A horizontal ray, coinciding with the negative x axis, with its rightmost point at B. A vertical ray, coinciding with the negative y axis, with its topmost point at B. A vertical ray with its topmost point H at (1,-0.34459). A diagonal line l containing points Z, H, and G in order from lower left to upper right, and intersecting the y axis. The line segment BZ is perpendicular to line l.)

We wish to calculate an exact expression for the *a* value of this corner geometry. Although illustrated for an outward-folding corner, the calculation is identical for an alternate-folding corner.

Theorem 3. In a two-facet corner with a rib fragment altitude of 1, the corner's *a* value is $\frac{2+\sqrt{2}+\sqrt{6+12\sqrt{2}}}{4}$.

Proof. [14] To use the point to line distance formula in [4]

$$\text{distance} = \frac{|k + mx_0 - y_0|}{\sqrt{1+m^2}}$$

we find the equation of the line determined by *GH* in the slope-intercept form $y = mx + k$

The slope of *GH* is $\frac{d}{a-1}$, and substituting the coordinates of $G\ (a, 1)$ to find *k* from the equation of a line

$$k = 1 - \frac{da}{a-1}$$

Equating the point to line distance formula to 1 and filling in the values *m* and *k* we get

$$1 = \frac{|k|}{\sqrt{1+m^2}} = \frac{\left|1-\frac{da}{a-1}\right|}{\sqrt{1+\left(\frac{d}{a-1}\right)^2}} = \frac{\left|1-\frac{da}{a-1}\right|}{\sqrt{1+\frac{d^2}{(a-1)^2}}}$$

Squaring both sides and multiplying by the resulting denominator gives

$$\left(1 - \frac{da}{a-1}\right)^2 = 1 + \frac{d^2}{(a-1)^2}$$

For ease of convenience in algebraic transformation, let $u = a - 1$. Then

$$\left(1 - \frac{d(u+1)}{u}\right)^2 = 1 + \frac{d^2}{u^2} \quad \Rightarrow \quad \frac{(u-du-d)^2}{u^2} = \frac{u^2+d^2}{u^2} \quad \Rightarrow \quad (u - du - d)^2 = u^2 + d^2$$

Expanding the left side gives

$$u^2 - du^2 - du - du^2 + d^2u^2 + d^2u - du + d^2u + d^2 = u^2 + d^2$$

Simplifying and collecting like terms and factoring gives

$$-du^2 - du - du^2 + d^2u^2 + d^2u - du + d^2u = 0$$

$$(-2d + d^2)u^2 + (-2d + 2d^2)u = 0$$

From here we get two solutions: $u = 0$ and $u = \frac{2d-2d^2}{-2d+d^2} = \frac{2d(1-d)}{d(-2+d)} = \frac{2(1-d)}{-2+d}$.

[14] Credit: Algebra calculations assisted by Google Gemini, Google, September 7&8, 2026. https://www.google.com/

From $u = 0$ we get $a = 1$, which we reject because it causes *GH* and *HJ* (cf. Figure 2) to be collinear, producing a degenerate two-facet corner with a ridiculously high parametrical error which seems no better than a conventional reservoir construction.

As for the second solution which gives $u(-2 + d) = 2(1 - d)$, we substitute $u = a - 1$ and $d = a - \frac{1}{\sqrt{2}}$ to get

$$(a-1)\left(-2 + a - \frac{1}{\sqrt{2}}\right) = 2\left(1 - \left(a - \frac{1}{\sqrt{2}}\right)\right) = 2\left(1 - a + \frac{1}{\sqrt{2}}\right)$$

Expanding and collecting like terms

$$-2a + a^2 - a\frac{1}{\sqrt{2}} + 2 - a + \frac{1}{\sqrt{2}} = 2 - 2a + \sqrt{2} \quad \Rightarrow \quad a^2 - a\frac{1}{\sqrt{2}} - a + \frac{1}{\sqrt{2}} = \sqrt{2}$$

produces a quadratic in *a*

$$a^2 - \left(1 + \frac{1}{\sqrt{2}}\right)a + \left(\frac{1}{\sqrt{2}} - \sqrt{2}\right) = 0$$

Applying the quadratic formula, we get the two values for *a*: the positive $a^+ = \frac{2+\sqrt{2}+\sqrt{6+12\sqrt{2}}}{4}$ and the negative $a^- = \frac{2+\sqrt{2}-\sqrt{6+12\sqrt{2}}}{4}$.

But since *a* is interpreted as a length, ang lengths are always positive, we can dismiss the negative value and accept the positive value as a_c. ■

The negative root does have an analytic geometric interpretation: These two roots induce the same corner geometry, reflected across the line $y = x$. That is to say, the positive value produces the configuration shown in Figure 24 where point *G* has a positive *x* coordinate. The negative value produces a similar configuration, but with points *G* and *H* reflected across the line $y = x$, and with their labels swapped. This produces an identically-shaped 'beak' contiguous with a rib, but oriented vertically, rather than horizontally. Said another way:

When $d = a^+ - \frac{1}{\sqrt{2}}$, point *G* has coordinates $(a^+, 1)$ and point *H* has coordinates $(1, a^-)$

When $d = a^- - \frac{1}{\sqrt{2}}$, point *H* has coordinates $(1, a^+)$ and point *G* has coordinates $(a^-, 1)$

Since $d_c = a_c - \frac{1}{\sqrt{2}}$, d_c has a beautifully symmetrical exact form $\frac{2-\sqrt{2}+\sqrt{6+12\sqrt{2}}}{4}$

Multiplying a_c and d_c produces the surprising result[15] that $a_c \cdot d_c = \frac{2+3\sqrt{2}+\sqrt{6+12\sqrt{2}}}{4}$

[15] Calculation credit: Wolfram Alpha LLC. 2026. Wolfram|Alpha. http://www.wolframalpha.com (access August 11, 2026)

16. Mathematical Background for Analysis of *GH* Length Variation

We begin by defining a function $GH(a, d, h)$ as the length of *GH* with an *a* value *a*, a *d* value of *d* at inflation height $h \in \left[0, \frac{1}{\sqrt{2}}\right]$.

Lemma 7. Over the inflation range $h \in [0, \frac{1}{\sqrt{2}}]$ the length *GH* in a two-facet corner characterized by a value *a* and d value *d* is

$$GH(a, d, h) = \sqrt{\left(a - \sqrt{1 - h^2}\right)^2 + \left(\sqrt{1 - h^2} + d - 1\right)^2}$$

Proof. In Figure 2, Figure 7, and Figure 9, choose point *B* as the origin. As *h* increases from 0 to $\frac{1}{\sqrt{2}}$, both point *G*'s y-coordinate and point *H*'s x-coordinate decrease. Thus

$$G = \left(a, \sqrt{1 - h^2}\right)$$

And

$$H = \left(\sqrt{1 - h^2}, -(d - 1)\right) = \left(\sqrt{1 - h^2}, -d + 1\right)$$

From the Cartesian distance formula, we find

$$\text{length } GH = \sqrt{(G_x - H_x)^2 + \left(G_y - H_y\right)^2}$$
$$= \sqrt{\left(\text{a} - \sqrt{1 - \text{h}^2}\right)^2 + \left(\sqrt{1 - \text{h}^2} - (-\text{d} + 1)\right)^2}$$

■

Notice that $GH(a, d, h)$ is well-defined for all *a*, *d*, and for $h \leq 1$. It suffices to assume $a > 1$ to make it non-zero. For clarity, we define another overloaded function $GH(a, h)$ as the length of *GH* with an *a* value *a*, a d value of $a - \frac{1}{\sqrt{2}}$ at inflation height *h* over the domain $0 \leq h \leq \frac{1}{\sqrt{2}}$.

$$GH(a, h) = \sqrt{\left(a - \sqrt{1 - h^2}\right)^2 + \left(\sqrt{1 - h^2} + a - \frac{1}{\sqrt{2}} - 1\right)^2}$$

To address the optimality of $d = a - \frac{1}{\sqrt{2}}$ we will consider how the parametrical error changes when we adjust *d* ($= a - \frac{1}{\sqrt{2}}$) by a change δ (lowercase delta). So, we define a function $GH\delta(a, h, \delta)$ as the length of *GH* with an *a* value *a* and *d* value $a - \frac{1}{\sqrt{2}} + \delta$, at inflation height $h \in \left[0, \frac{1}{\sqrt{2}}\right]$. Thus, from Lemma 7,

$$GH\delta(a, h, \delta) = \sqrt{\left(a - \sqrt{1 - h^2}\right)^2 + \left(\sqrt{1 - h^2} + \left(a - \frac{1}{\sqrt{2}} + \delta\right) - 1\right)^2}$$

That is to say, $GH(a, h) = GH\delta(a, h, 0)$

Lemma 8. For any *a* and δ,

$$GH\delta(a,0,\delta) = \sqrt{2a^2 + 2a\delta + \delta^2 - (2+\sqrt{2})a - \delta\sqrt{2} + \frac{3}{2}}$$

Proof. From the definition, $GH\delta(a,h,\delta)$

$$= \sqrt{\left(a - \sqrt{1-0^2}\right)^2 + \left(\sqrt{1-0^2} + a - \frac{1}{\sqrt{2}} + \delta - 1\right)^2}$$
$$= \sqrt{(a-1)^2 + \left(\ a\ +\ \delta\ - \frac{1}{\sqrt{2}}\ \right)^2}$$
$$= \sqrt{2a^2 + 2a\delta + \delta^2 - (2+\sqrt{2})a - \delta\sqrt{2} + \frac{3}{2}}$$

■

Lemma 9. For any a and δ,

$$GH\delta\left(a, \frac{1}{\sqrt{2}}, \delta\right) = \sqrt{2a^2 - 2a - a\sqrt{2} + \delta^2 - 2\delta + 2\delta a + \frac{3}{2}}$$

Proof. [16] Let $X = \sqrt{1 - \left(\frac{1}{\sqrt{2}}\right)^2} = \frac{1}{\sqrt{2}}$.

From the definition of $GH\delta(a,h,\delta)$,

$$GH\delta\left(a, \tfrac{1}{\sqrt{2}}, \delta\right) = \sqrt{(a-X)^2 + \left(X + a - \tfrac{1}{\sqrt{2}} + \delta - 1\right)^2}$$

$$= \sqrt{a^2 - 2aX + X^2 + (a+\delta-1)^2}$$
$$= \sqrt{a^2 - 2aX + X^2 + a^2 + a\delta - a + a\delta + \delta^2 - \delta - a - \delta + 1}$$
$$= \sqrt{a^2 - \frac{2}{\sqrt{2}}a + \frac{1}{2} + a^2 + a\delta - a + a\delta + \delta^2 - \delta - a - \delta + 1}$$
$$= \sqrt{2a^2 - \sqrt{2}a + 2a\delta - 2a + \delta^2 - 2\delta + \frac{3}{2}}$$

Rearranging the terms and factors gives the form shown above. ■

Lemma 10. Independent of a value, if $\delta \geq 0$, then $GH\delta\left(a, \frac{1}{\sqrt{2}}, \delta\right) \leq GH\delta(a,0,\delta)$.

Proof. [17] From Lemma 8 and Lemma 9 we have

$$GH\delta(a,0,\delta) = \sqrt{2a^2 + 2a\delta + \delta^2 - (2+\sqrt{2})a - \delta\sqrt{2} + \frac{3}{2}}$$
$$GH\delta\left(a, \tfrac{1}{\sqrt{2}}, \delta\right) = \sqrt{2a^2 - 2a - a\sqrt{2} + \delta^2 - 2\delta + 2\delta a + \tfrac{3}{2}}$$

[16] Credit: Algebra calculations assisted by Google Gemini, Google, August 20&22, 2026. https://www.google.com/
[17] Credit: Algebra calculations assisted by Google Gemini, Google, August 20&22, 2026. https://www.google.com/

If $x^2 - y^2 \leq 0$ then $x^2 \leq y^2$. For non-negative numbers (e.g. lengths) this implies that $x \leq y$

Calculating $GH\delta\left(a, \frac{1}{\sqrt{2}}, \delta\right)^2 - GH\delta(a,0,\delta)^2$,

$$
\begin{aligned}
&= \left(2a^2 - 2a - a\sqrt{2} + \delta^2 - 2\delta + 2\delta a + \frac{3}{2}\right) - \left(2a^2 + 2a\delta + \delta^2 - \left(2+\sqrt{2}\right)a - \delta\sqrt{2} + \frac{3}{2}\right) \\
&= 2a^2 - 2a - a\sqrt{2} + \delta^2 - 2\delta + 2\delta a + \frac{3}{2} - 2a^2 - 2a\delta - \delta^2 + \left(2+\sqrt{2}\right)a + \delta\sqrt{2} - \frac{3}{2} \\
&= -2a - a\sqrt{2} - 2\delta + 2a + a\sqrt{2} + \delta\sqrt{2} \\
&= -2\delta + \delta\sqrt{2}
\end{aligned}
$$

which is $\delta\left(\sqrt{2}-2\right)$

Since $\left(\sqrt{2}-2\right) < 0$ and we assumed $\delta \geq 0$, then $GH\delta\left(a, \frac{1}{\sqrt{2}}, \delta\right) \leq GH\delta(a,0,\delta)$ ■

Corollary 2. Independent of *a* value, if $\delta \leq 0$, then $GH\delta\left(a, \frac{1}{\sqrt{2}}, \delta\right) \geq GH\delta(a,0,\delta)$.

Proof. We construct $GH\delta\left(a, \frac{1}{\sqrt{2}}, \delta\right)$ and $GH\delta(a,0,\delta)$ and calculate

$$GH\delta\left(a, \frac{1}{\sqrt{2}}, \delta\right)^2 - GH\delta(a,0,\delta)^2 = \delta\left(\sqrt{2}-2\right)$$

exactly as in Lemma 10.

If $x^2 - y^2 \geq 0$ then $x^2 \geq y^2$. For positive numbers (e.g. lengths) this implies that $x \geq y$

Since $\left(\sqrt{2}-2\right) < 0$ and we assumed $\delta \leq 0$, then $GH\delta\left(a, \frac{1}{\sqrt{2}}, \delta\right) \geq GH\delta(a,0,\delta)$. ■

Remark: Notice that when $\delta < 0$, the proof of Corollary 2 would indicate that $GH\delta\left(a, \frac{1}{\sqrt{2}}, \delta\right) > GH\delta(a,0,\delta)$

Corollary 3. Independent of *a* value, when $\delta = 0$,

$$GH(a,0) = GH\delta(a,0,0) = GH\delta\left(a, \frac{1}{\sqrt{2}}, 0\right) = GH\left(a, \frac{1}{\sqrt{2}}\right).$$

Proof. From the definitions of $GH(a,h)$ and $GH\delta(a,h,\delta)$, we see that $GH(a,h)$ is exactly $GH\delta(a,h,\delta)$ when $\delta = 0$.

We know from Corollary 2 and Lemma 10, that independent of *a* value:

$GH\delta\left(a, \frac{1}{\sqrt{2}}, 0\right) \geq GH\delta(a,0,0)$ and $GH\delta\left(a, \frac{1}{\sqrt{2}}, 0\right) \leq GH\delta(a,0,0)$

So it must be true that $GH(a,0) = GH\delta(a,0,0) = GH\delta\left(a, \frac{1}{\sqrt{2}}, 0\right) = GH\left(a, \frac{1}{\sqrt{2}}\right)$, when combined with the definition of $GH(a,h)$. ■

Lemma 11. Over $-\frac{1}{\sqrt{2}} \leq h \leq \frac{1}{\sqrt{2}}$, and independent of *a* value, $GH\delta(a,h,\delta)$ has a minimum at

$$h_K = \frac{1}{2} \cdot \sqrt{\frac{5}{2} - \sqrt{2} + \left(2 + \sqrt{2}\right)\delta - \delta^2}$$

and, if $h_K \neq 0$, a maximum at $h = 0$.

Proof. [18] Taking the partial derivative of $GH\delta(a, h, \delta)$ with respect to *h* we find:

$$\frac{h\left(1 + \frac{1}{\sqrt{2}} - \delta - 2\sqrt{1 - h^2}\right)}{\sqrt{1 - h^2} \cdot GH\delta(a, h, \delta)}$$

Since we are looking for places where the derivative of $GH\delta(a, h, \delta)$ is 0 indicating a critical point (e.g. minimum or maximum), and since a rational expression is zero when its numerator is zero (and its denominator is non-zero, which it is in the domain $-\frac{1}{\sqrt{2}} \leq h \leq \frac{1}{\sqrt{2}}$- we can set the numerator equal to 0 and solve for h to find three critical points in $[-\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}]$: 0, h_K, and h'_K.

$$h_K = \frac{1}{2} \cdot \sqrt{\frac{5}{2} - \sqrt{2} + \left(2 + \sqrt{2}\right)\delta - \delta^2}$$
$$h'_K = -h_K$$

In the process of solving $1 + \frac{1}{\sqrt{2}} - \delta - 2\sqrt{1 - h^2} = 0$ to find h_K, we found the intermediate step $1 + \frac{1}{\sqrt{2}} - \delta = 2\sqrt{1 - h^2}$. Since we eventually find that h_K satisfies this equation, we know

(4) $1 + \frac{1}{\sqrt{2}} - \delta = 2\sqrt{1 - h_K^2}$

h_K is a non-negative critical point while h'_K is its non-positive reflection across the *y*-axis, due to the fact that $GH\delta(a, h, \delta)$ is an even function when h is taken as its independent variable.

At a minimum point, the function's slope is zero (which is true of $GH\delta(a, h_K, \delta)$ by definition of a critical point), and the function's slopes to the left and right are downward toward the right and upward toward the right, respectively.

In the partial derivative above we see that both factors of the denominator are always positive, so the sign of the slope expressed by the derivative is the sign of the derivative's numerator alone:

$$N(h, \delta) = h\left(1 + \frac{1}{\sqrt{2}} - \delta - 2\sqrt{1 - h^2}\right)$$

Suppose we nudge h_K slightly to the right and left by a positive or negative infinitesimal ε (lowercase epsilon), respectively.

Substituting nudged $h = h_K + \varepsilon$ and $1 + \frac{1}{\sqrt{2}} - \delta = 2\sqrt{1 - h_K^2}$ into the numerator, we can express the numerator as:

$$N(h, \delta) = (h_K + \varepsilon)\left(2\sqrt{1 - h_K^2} - 2\sqrt{1 - (h_K + \varepsilon)^2}\right) = 2(h_K + \varepsilon)\left(\sqrt{1 - h_K^2} - \sqrt{1 - (h_K + \varepsilon)^2}\right)$$

[18] Credit: Proof review, and algebra and derivative calculations assisted, by Google Gemini, Google, August 20, 23 and September 6, 2026. https://www.google.com/ Algebra calculations assisted by Wolfram Alpha LLC. 2026. Wolfram|Alpha. http://www.wolframalpha.com (access August 23, 2026).

Case I: When $\varepsilon > 0$, both the $(h_K + \varepsilon)$ term and the $\sqrt{1 - h_K^2} - \sqrt{1 - (h_K + \varepsilon)^2}$ term are positive. Thus, the numerator, and the derivative as a whole, is positive. This indicates that at $h_K + \varepsilon$ the curve of $GH\delta(a, h, \delta)$ to the right of h_K is sloping upward to the right.

Case II: When $\varepsilon < 0$, to examine the curve of $GH\delta(a, h, \delta)$ to the left of h_K, we have to consider two cases: $h_K = 0$ and $h_K > 0$

Case IIa: When $h_K = 0$, the numerator becomes $2(\varepsilon)\left(1 - \sqrt{1 - \varepsilon^2}\right)$ which is negative, indicating that the curve of $GH\delta(a, h, \delta)$ to the left of h_K (i.e. to the left of 0) is sloping downward toward the right.

Case IIb: When $h_K > 0$, we are only concerned with the region of $GH\delta(a, h, \delta)$ between h_K and 0. So $0 < h_K + \varepsilon < h_K$.

When $\varepsilon < 0$, the $(h_K + \varepsilon)$ term is still positive, but because $\sqrt{1 - (h_K + \varepsilon)^2}$ is larger than $\sqrt{1 - h_K^2}$ the $\sqrt{1 - h_K^2} - \sqrt{1 - (h_K + \varepsilon)^2}$ term is negative. Thus, the numerator, and the derivative as a whole, is negative. This indicates that at $h_K + \varepsilon$ and $\varepsilon < 0$ the curve of $GH\delta(a, h, \delta)$ to the left of h_K is sloping downward toward the right.

Combining both cases IIa and IIb: At $h_K + \varepsilon$ when $\varepsilon < 0$ the curve of $GH\delta(a, h, \delta)$ to the left of h_K is sloping downward toward the right.

Finally, combining cases I and II, we conclude that $GH\delta(a, h_K, \delta)$ is a minimum.

If $h_K \neq 0$ then $GH\delta(a, 0, \delta)$ is the maximum since $h = 0$ is the only critical point between minimum critical points h'_K and h_K. If $h_K = 0$, then critical points 0, h'_K and h_K all coincide and $GH\delta(a, 0, \delta) = GH\delta(a, h_K, \delta)$. ■

Note that *a* dropped out of the numerator in the partial derivative with respect to *h*, and thus the *h* values for which length *GH* is at its minimum and maximum are independent of the chosen *a* value.

Corollary 4. Over $0 \leq h \leq \frac{1}{\sqrt{2}}$, and independent of *a* value, $GH(a, h)$ has a single minimum value at $h_K = \frac{1}{2} \cdot \sqrt{\frac{5}{2} - \sqrt{2}}$

Proof. This follows directly from Lemma 11 where $\delta = 0$ and by definitions of $GH(a, h)$ and $GH\delta(a, h, \delta)$ ■

Recall that in Section 5 we defined a parametrical error function, notated as $\omega(a)$, for a given a value as the difference between the maximum and minimum lengths of *GH* when $d = a - \frac{1}{\sqrt{2}}$ over the domain $0 \leq h \leq \frac{1}{\sqrt{2}}$

Thus, defined formally,

$$\omega(a) = \max_{0 \leq h \leq \frac{1}{\sqrt{2}}} GH(a, h) \quad - \quad \min_{0 \leq h \leq \frac{1}{\sqrt{2}}} GH(a, h)$$

Figure 25 shows a graph of the function $GH(a,h)$ over the domain $0 \le h \le \frac{1}{\sqrt{2}}$ and summarizes the findings thus far.

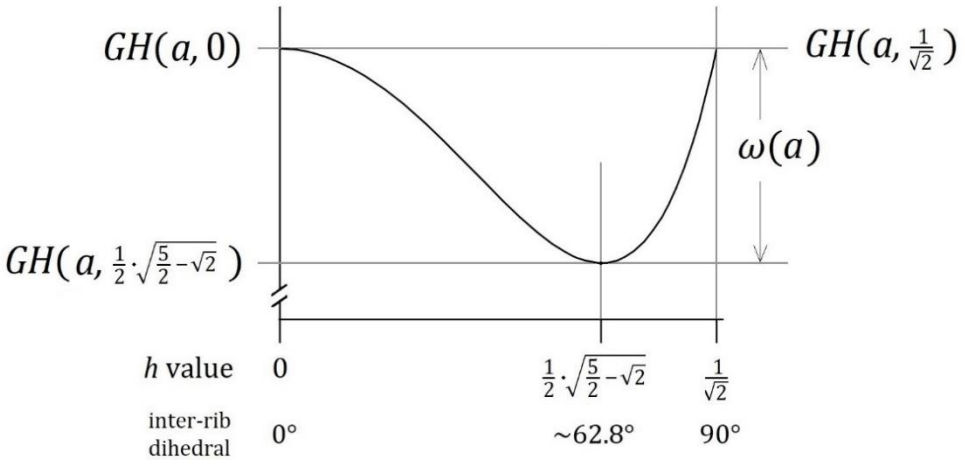


Figure 25

Lemma 12. Over $0 \le h \le \frac{1}{\sqrt{2}}$, and independent of *a* value, $GH\delta(a,h_K,\delta)$ is well-defined when δ is bounded as $\frac{\sqrt{2}}{2} - 1 \le \delta \le 1 - \frac{\sqrt{2}}{2} < \frac{2+2\sqrt{2}}{2}$.

Proof. [19] By the bounds on *h*, we know h_K must lie in $0 \le h_K \le \frac{1}{\sqrt{2}}$ and we know from Lemma 11 that $h_K = \frac{1}{2}\cdot\sqrt{\frac{5}{2} - \sqrt{2} + (2+\sqrt{2})\delta - \delta^2}$

Characterizing the set of acceptable δ values:

Solving $0 \le \frac{1}{2}\cdot\sqrt{\frac{5}{2} - \sqrt{2} + (2+\sqrt{2})\delta - \delta^2}$ we find

$\frac{\sqrt{2}}{2} - 1 \le \delta \le 3 + \frac{\sqrt{2}}{2}$ (~ -0.2928 to ~ 3.7071)

Adding the constraint that $\frac{1}{2}\cdot\sqrt{\frac{5}{2} - \sqrt{2} + (2+\sqrt{2})\delta - \delta^2} \le \frac{1}{\sqrt{2}}$ we can see that additionally

$\delta \le 1 - \frac{\sqrt{2}}{2}$ (~ 0.2928) and $d \ge 1 + \frac{3\sqrt{2}}{2}$ (~ 3.1213)

Thus, the δ values that satisfy $0 \le \frac{1}{2}\cdot\sqrt{\frac{5}{2} - \sqrt{2} + (2+\sqrt{2})\delta - \delta^2} \le \frac{1}{\sqrt{2}}$ lie in two disjoint sets:

$\frac{\sqrt{2}}{2} - 1 \le \delta \le 1 - \frac{\sqrt{2}}{2}$ and $1 + \frac{3\sqrt{2}}{2} \le \delta \le 3 + \frac{\sqrt{2}}{2}$

Note that the right side region is an extraneous result, since it would make the value of the root function of equation (4) of Lemma 11 negative.

[19] Credit: Proof phrasing assistance and algebra calculations assisted by Google Gemini, Google, August 22 and September 5, 2026. https://www.google.com/ Plotting, and algebra and derivative calculations assisted by Wolfram Alpha LLC. 2026. Wolfram|Alpha. http://www.wolframalpha.com (access August 22 and September 5, 2026).

As for the left side region, we wish to show that $[0,\frac{1}{\sqrt{2}}]$ is the codomain of a function $f(\delta) = h_K = \frac{1}{2}\cdot\sqrt{\frac{5}{2} - \sqrt{2} + (2+\sqrt{2})\delta - \delta^2}$ even if δ be restricted to $[\frac{\sqrt{2}}{2} - 1, 1 - \frac{\sqrt{2}}{2}]$. That is, we wish to show that f is a surjection from $[\frac{\sqrt{2}}{2} - 1, 1 - \frac{\sqrt{2}}{2}]$ onto $[0,\frac{1}{\sqrt{2}}]$.

We can verify by substitution that $f\left(\frac{\sqrt{2}}{2} - 1\right) = 0$ and $f\left(1 - \frac{\sqrt{2}}{2}\right) = \frac{1}{\sqrt{2}}$.

The function f is differentiable as:

$$f' = \frac{-2\delta + \sqrt{2} + 2}{8f}$$

and is increasing everywhere in $[\frac{\sqrt{2}}{2} - 1, 1 - \frac{\sqrt{2}}{2}]$. Since f is differentiable over $\left[\frac{\sqrt{2}}{2} - 1, 1 - \frac{\sqrt{2}}{2}\right]$ we know f is continuous over $\left[\frac{\sqrt{2}}{2} - 1, 1 - \frac{\sqrt{2}}{2}\right]$. Now, let y be an arbitrary value in the codomain $\left[0,\frac{1}{\sqrt{2}}\right]$ of f. By the Intermediate Value Theorem [6], since $f\left(\frac{\sqrt{2}}{2} - 1\right) < y < f\left(1 - \frac{\sqrt{2}}{2}\right)$, there exists some δ between $\frac{\sqrt{2}}{2} - 1$ and $1 - \frac{\sqrt{2}}{2}$ such that $f(\delta) = y$. Because every element in the codomain possesses at least one preimage under f, the function f is a surjection when its domain is $\frac{\sqrt{2}}{2} - 1 \le \delta \le 1 - \frac{\sqrt{2}}{2}$. This permits us to simplify our analysis of the behavior of $GH\delta(a, h_K, \delta)$ where $h \in \left[0,\frac{1}{\sqrt{2}}\right]$ by confining our analysis to $\delta \in \left[\frac{\sqrt{2}}{2} - 1, 1 - \frac{\sqrt{2}}{2}\right]$.

Moreover, everywhere within the left side region, $\delta \le 1 - \frac{\sqrt{2}}{2} < \frac{2+2\sqrt{2}}{2}$. ■

Lemma 13. Over $0 \le h \le \frac{1}{\sqrt{2}}$, and independent of a value,

$$GH\delta(a, h_K, \delta) = \sqrt{2}\left|a + \frac{\delta}{2} - \frac{2+\sqrt{2}}{4}\right|$$

But when $a > 1$,

$$GH\delta(a, h_K, \delta) = \sqrt{2}\left(a + \frac{\delta}{2} - \frac{2+\sqrt{2}}{4}\right)$$

Proof. [20] Notice from Lemma 11 that $h_K^2 = \frac{1}{4}\cdot\left(\frac{5}{2} - \sqrt{2} + (2+\sqrt{2})\delta - \delta^2\right)$

And

$$\begin{aligned} 1 - h_K^2 &= 1 - \frac{5}{8} + \frac{\sqrt{2}}{4} - \delta\frac{2+\sqrt{2}}{4} + \frac{\delta^2}{4} \\ &= \frac{\delta^2}{4} - \delta\frac{2+\sqrt{2}}{4} + \frac{3}{8} + \frac{\sqrt{2}}{4} \\ &= \frac{1}{4}\cdot\left(\delta^2 - \delta(2+\sqrt{2}) + \left(\frac{3}{2} + \sqrt{2}\right)\right) \end{aligned}$$

[20] Credit: Algebra calculations assisted by Google Gemini, Google, August 22, 2026. https://www.google.com/ Algebra calculations assisted by Wolfram Alpha LLC. 2026. Wolfram|Alpha. http://www.wolframalpha.com (access August 22, 2026).

Notice that $\left(\frac{3}{2}+\sqrt{2}\right)=\frac{6+4\sqrt{2}}{4}=\left(\frac{2+\sqrt{2}}{2}\right)^2$ so

$1-h_K^2=\frac{1}{4}\cdot\left(\delta-\frac{2+\sqrt{2}}{2}\right)^2$ and $\sqrt{1-h_K^2}=\frac{1}{2}\cdot\left|\delta-\frac{2+\sqrt{2}}{2}\right|$

By Lemma 12, $\delta<\frac{2+2\sqrt{2}}{2}$. So, we take the negative case of the absolute value

Now let $Y=\sqrt{1-h_K^2}=-\frac{\delta}{2}+\frac{2+\sqrt{2}}{4}$

We can express $GH\delta(a,h_K,\delta)$ by substituting Y and $d=a-\frac{1}{\sqrt{2}}+\delta$ into the expression for $GH\delta(a,h,\delta)$ from Lemma 7

$$GH\delta(a,h_K,\delta)=\sqrt{(a-Y)^2+\left(Y+a-\frac{1}{\sqrt{2}}+\delta-1\right)^2}$$

Let $C=\delta-1-\frac{1}{\sqrt{2}}$. then

$$\begin{aligned}GH\delta(a,h_K,\delta)&=\sqrt{(a-Y)^2+(a+Y+C)^2}\\&=\sqrt{2a^2+2Y^2+2aC+2YC+C^2}\end{aligned}$$

Recall that $Y=\sqrt{1-h_K^2}=-\frac{\delta}{2}+\frac{2+\sqrt{2}}{4}$

So $Y=-\frac{\delta}{2}+\frac{2+2\sqrt{2}}{4}=-\frac{\delta}{2}+\frac{1}{2}+\frac{\sqrt{2}}{2}=-\frac{1}{2}\left(\delta-1-\frac{1}{\sqrt{2}}\right)=-\frac{1}{2}C$

Implying that $C=-2Y$, and thus

$$\begin{aligned}GH\delta(a,h_K,\delta)&=\sqrt{2a^2+2Y^2-4aY-4YY+4YY}\\&=\sqrt{2a^2-4aY+2Y^2}\\&=\sqrt{2(a-Y)^2}\\&=\sqrt{2}|a-Y|\\&=\sqrt{2}\left|a-\left(-\frac{\delta}{2}+\frac{2+\sqrt{2}}{4}\right)\right|\\&=\sqrt{2}\left|a+\frac{\delta}{2}-\frac{2+\sqrt{2}}{4}\right|\end{aligned}$$

When $a>1$, $a-\sqrt{1-h_K^2}$ is always positive, thus $a-Y$ is positive, so from above

$$GH\delta(a,h_K,\delta)=\sqrt{2}|a-Y|=\sqrt{2}\left(a-\left(-\frac{\delta}{2}+\frac{2+\sqrt{2}}{4}\right)\right)=\sqrt{2}\left(a+\frac{\delta}{2}-\frac{2+\sqrt{2}}{4}\right)$$

■

17. Characteristics of $\omega(a)$

In this section, we prove some properties of $\omega(a)$.

Theorem 4. $\omega(a) = \sqrt{(a-1)^2 + \left(a - \frac{1}{\sqrt{2}}\right)^2} - \sqrt{2}\left|a - \frac{2+\sqrt{2}}{4}\right|$

Proof. From Lemma 11 when $\delta = 0$ we know that $GH\delta(a, 0,0)$ is the maximum and $GH\delta(a, h_K, 0)$ is the minimum.

From Lemma 8 we know

$$GH\delta(a,0,0) = \sqrt{2a^2 - \left(2+\sqrt{2}\right)a + \frac{3}{2}}$$
$$= \sqrt{a^2 - 2a + 1 + a^2 - 2a\left(\frac{1}{\sqrt{2}}\right) + \frac{1}{2}}$$
$$= \sqrt{(a-1)^2 + \left(a - \frac{1}{\sqrt{2}}\right)^2}$$

and from Lemma 13 we know

$$GH\delta(a, h_K, 0) = \sqrt{2}\left|a - \frac{2+\sqrt{2}}{4}\right|$$

Since $GH(a,h) = GH\delta(a,h,0)$ we get the result from the definition of $\omega(a)$:

$$\omega(a) = \max_{0 \le h \le \frac{1}{\sqrt{2}}} GH\delta(a,h,0) - \min_{0 \le h \le \frac{1}{\sqrt{2}}} GH\delta(a,h,0)$$

$$= GH\delta(a,0,0) - GH\delta(a,h_K,0)$$

■

Corollary 5: if $y > x > 1$ then $\omega(x) > \omega(y)$

Proof. [21] From Theorem 4 we know

$$\omega(a) = GH\delta(a,0,0) - \sqrt{2}\left|a - \frac{2+\sqrt{2}}{4}\right|$$

And in particular, when $a > 1$

$$\omega(a) = GH\delta(a,0,0) - \sqrt{2}\left(a - \frac{2+\sqrt{2}}{4}\right)$$

where $GH\delta(a,0,0) = \sqrt{2a^2 - \left(2+\sqrt{2}\right)a + \frac{3}{2}}$

[21] Credit: Proof comments and checking provided by Google Gemini, Google, August 30 and September 13, 2026. https://www.google.com/ Calculating derivative and algebra assistance by Wolfram Alpha LLC. 2026. Wolfram|Alpha. http://www.wolframalpha.com (access August 29, 2026).

We observe that $\omega(a)$ is both continuous and differentiable over $\mathbb{R}$ (real numbers) over this restricted domain $a > 1$, satisfying the Mean Value Theorem [5].

We differentiate this latter version of $\omega(a)$ and combine terms over a common denominator to obtain

$$\omega'(a) = \frac{\left(4a - \sqrt{2} - 2\right) - 2\sqrt{2} \cdot GH\delta(a, 0,0)}{2 \cdot GH\delta(a, 0,0)}$$

Notice that the derivative's denominator is always positive, because $GH\delta(a, 0,0)$ is always positive being the square root of the sum of two different values, as seen in the proof of Lemma 8. So, the sign of the numerator determines the sign of the entire derivative.

Consider the derivative's numerator. Let $X = \left(4a - \sqrt{2} - 2\right)$ and let $Y = 2\sqrt{2} \cdot GH\delta(a, 0,0)$

Squaring *X* and *Y* and substituting $GH\delta(a, 0,0)^2$ with its value from Lemma 101, we get

$$X^2 - Y^2 = \left(4a - \sqrt{2} - 2\right)^2 - 8\left(2a^2 - \left(2 + \sqrt{2}\right)a + \frac{3}{2}\right)$$

Expanding the first term gives $16a^2 - \sqrt{2}a * 8 - 16a + 6 + 4\sqrt{2}$

Expanding the second term gives $16a^2 - 16a - 8a\sqrt{2} + 12$

giving a constant difference of $4\sqrt{2} - 6$ (approximately -0.34) indicating that $X^2 - Y^2 < 0$ and $X^2 < Y^2$.

Y is non-negative because it represents a positive constant multiplied by a square. *X* is non-negative when $a > 1$. Since *X* and *Y* are non-negative, $X^2 < Y^2$ implies $X < Y$, that is $X - Y < 0$

But $X - Y$ is exactly the numerator of the derivative, and because it is negative, the entire derivative is negative over the domain $(1, \infty)$.

Now, $\omega(a)$ is continuous over $[x, y]$ and differentiable over (x, y) for any *x* and *y* such that $y > x > 1$. So we can apply the Mean Value Theorem [5] to find that there exists a value *c* such that $x < c < y$ so that

$$\omega'(c) = \frac{\omega(y) - \omega(x)}{y - x}$$

Since $\omega'(c)$ is negative for $c > 1$ and $y - x$ is positive, we conclude that $\omega(y) - \omega(x)$ must be negative, showing that $\omega(y) < \omega(x)$ ■

Corollary 6: $\lim_{a \to \infty} \omega(a) = 0$

Proof. [22] Since $a \gg \frac{2+\sqrt{2}}{4}$ and by Theorem 4 we can write

$$\omega(a) = \sqrt{(a-1)^2 + \left(a - \frac{1}{\sqrt{2}}\right)^2} - \sqrt{2}\left(a - \frac{2+\sqrt{2}}{4}\right)$$

[22] Credit: Calculating derivative and algebra assistance by Wolfram Alpha LLC. 2026. Wolfram|Alpha. http://www.wolframalpha.com (access August 30, 2026).

consider the conjugate of $\omega(a)$, which can be written in two equivalent forms, K_1 and K_2.

$$K_1 = \sqrt{(a-1)^2 + \left(a - \frac{1}{\sqrt{2}}\right)^2} + \sqrt{2}\left(a - \frac{2+\sqrt{2}}{4}\right)$$

$$K_2 = \sqrt{2a^2 - \left(2+\sqrt{2}\right)a + \frac{3}{2}} + \sqrt{2}\left(a - \frac{2+\sqrt{2}}{4}\right)$$

multiplying $\omega(a)$ by $\frac{K_2}{K_1}$ gives

$$\omega(a) = \frac{2a^2 - (2+\sqrt{2})a + \frac{3}{2} - 2\left(a - \frac{2+\sqrt{2}}{4}\right)^2}{K_1}$$

Expanding the numerator simplifies $\omega(a)$ to

$$\omega(a) = \frac{\frac{3}{4} - \frac{1}{\sqrt{2}}}{K_1}$$

Note that K_1 is entirely a composition of functions that are all increasing over $\left(\frac{2+\sqrt{2}}{4}, \infty\right)$ So as a approaches infinity, K_1 also approaches infinity. Thus, the limit of $\omega(a)$ as a approaches infinity is 0. ■

18. Sketching a Proof for the Optimality of $d = a - \frac{1}{\sqrt{2}}$ when $a > 1$

We now turn attention to sketching a proof for optimality of $d = a - \frac{1}{\sqrt{2}}$ for the case of $a > 1$. The author considers this case sufficient, since a values less than a_c are not practically useful, and Section 15 illustrated that an a value of 1 should be considered a degenerate case[23].

Stated formally: When $\delta = 0$, $d = a - \frac{1}{\sqrt{2}} + \delta$ gives the minimal parametrical error for all $a > 1$

Besides the constraint of $a > 1$, for practical purposes we know that since d represents a geometric length, that $d \geq 0$. Comparing the expressions for $GH(a, d, h)$ and $GH\delta(a, h, \delta)$ near the beginning of Section 16, we can observe that $a - \frac{1}{\sqrt{2}} + \delta \geq 0$, That is $\delta \geq \frac{1}{\sqrt{2}} - a$.

To begin, we define another parametrical error function, notated as $\omega\delta(a, \delta)$, for a given a and δ value as the difference between the maximum and minimum values of $GH\delta(a, h, \delta)$ over the domain $0 \leq h \leq \frac{1}{\sqrt{2}}$.

Thus,

$$\omega\delta(a, \delta) = \max_{0 \leq h \leq \frac{1}{\sqrt{2}}} GH\delta(a, h, \delta) - \min_{0 \leq h \leq \frac{1}{\sqrt{2}}} GH\delta(a, h, \delta)$$

Note then that $\omega(a) = \omega\delta(a, 0)$

[23] The author observed that this optimality appears to hold true for many real a values, both positive and negative, when he playfully plugged them into his numerical models. However, proving the result for $a \in \mathbb{R}$ would be a difficult exercise that neither the author nor many readers are likely to find delightful or satisfying.

We will develop a Mean Value Theorem argument to show that $\omega\delta(a,\delta)$ is minimized when $\delta = 0$ and thus $d = a - \frac{1}{\sqrt{2}}$ gives the smallest possible parametrical error $\omega\delta(a,\delta)$.

To do this, it is necessary to characterize $\omega\delta(a,\delta)$ as a piecewise function as follows:

$$\omega\delta(a,\delta) = \begin{cases} GH\delta(a,0,\delta) - GH\delta(a,h_K,\delta) & \text{when } 1 - \frac{\sqrt{2}}{2} \geq \delta \geq 0 \\ GH\delta\left(a,\frac{1}{\sqrt{2}},\delta\right) - GH\delta(a,h_K,\delta) & \text{when } \frac{\sqrt{2}}{2} - 1 \leq \delta \leq 0 \\ GH\delta\left(a,\frac{1}{\sqrt{2}},\delta\right) - GH\delta(a,0,\delta) & \text{when } \frac{\sqrt{2}}{2} - a \leq \delta < \frac{\sqrt{2}}{2} - 1 \\ GH\delta(a,0,\delta) - GH\delta\left(a,\frac{1}{\sqrt{2}},\delta\right) & \text{when } \delta > 1 - \frac{\sqrt{2}}{2} \end{cases}$$

Figure 26 shows an example graph of $GH\delta(a,h,\delta)$ where $a = 3$ and $1 - \frac{\sqrt{2}}{2} \geq \delta \geq 0$.

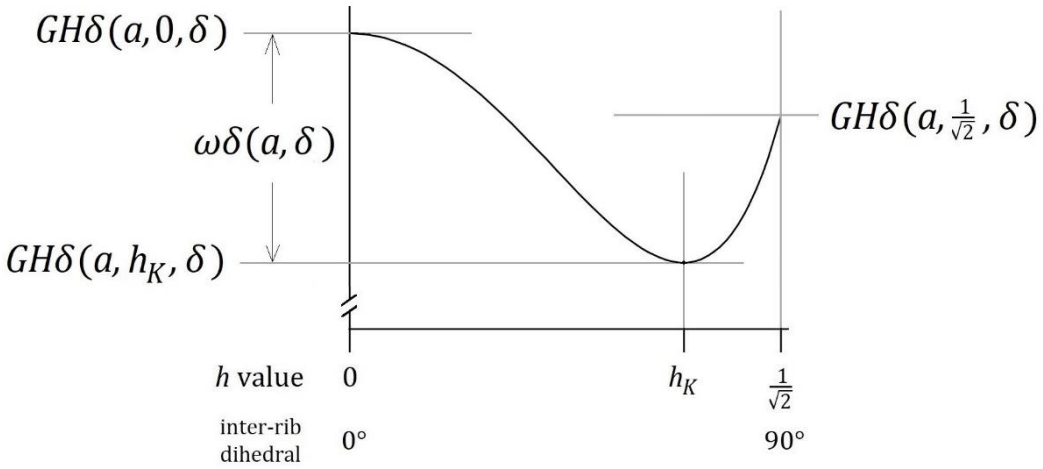


Figure 26

Figure 27 shows an example graph of $GH\delta(a,h,\delta)$ where $a = 3$ and $\frac{\sqrt{2}}{2} - 1 \leq \delta \leq 0$.

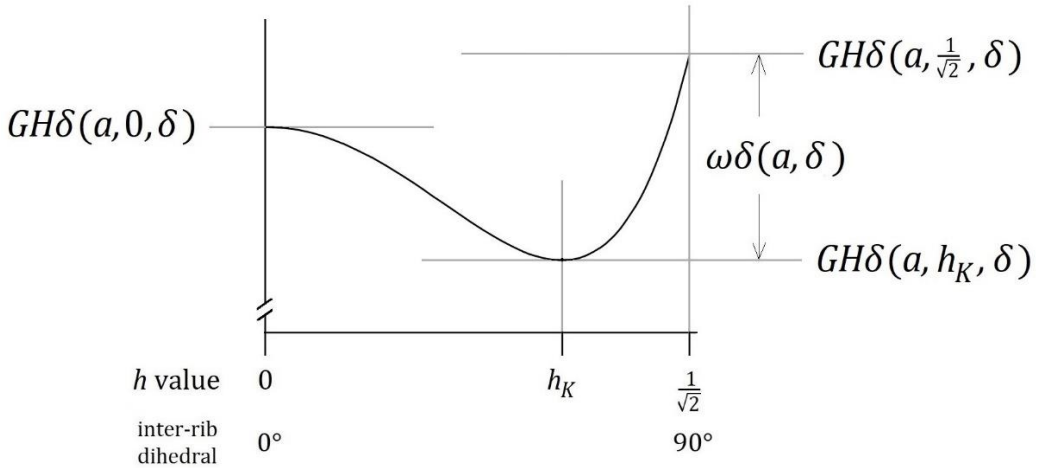


Figure 27

(Figure 25, above I n Section 16, shows an example graph of $GH\delta(a,h,\delta)$ where $a = 3$ and $\delta = 0$.)

Figure 28 shows am example graph of $\omega\delta(a,\delta)$ where $a = 3$.

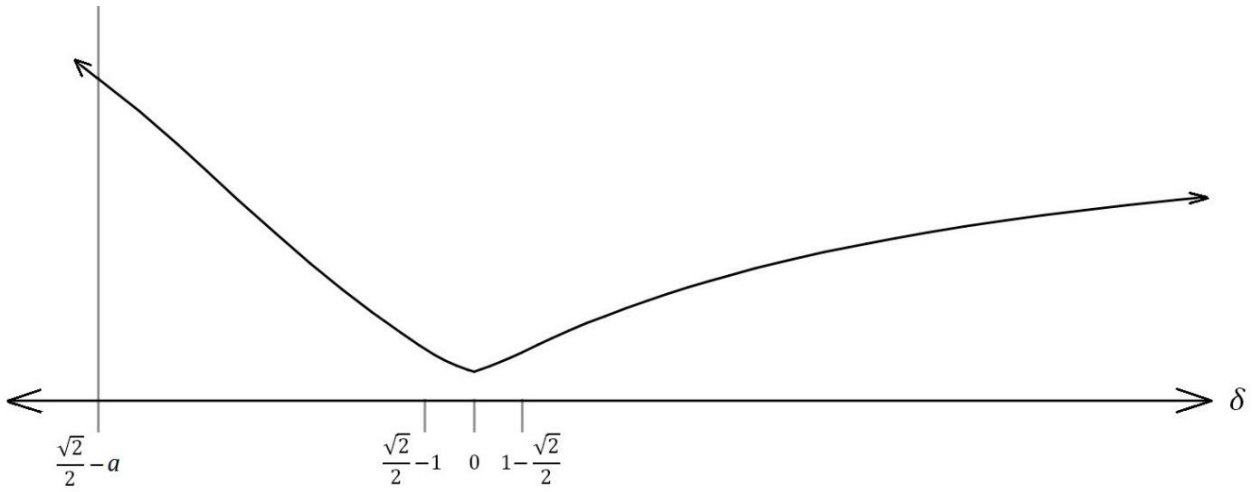


Figure 28

As δ approaches infinity, $\omega\delta(a,\delta)$ approaches $1-\frac{1}{\sqrt{2}}$.
Outside of $\delta \in \left[1-\frac{1}{\sqrt{2}}, 1-\frac{1}{\sqrt{2}}\right]$, as indicated by Lemma 12, $GH\delta(a,h_K,\delta)$ is not well-defined; $GH\delta(a,h,\delta)$ has no internal minimum point when $\delta \in \left[1-\frac{1}{\sqrt{2}}, 1-\frac{1}{\sqrt{2}}\right]$. But, Lemma 10 and Corollary 2 assure us this definition of $\omega\delta(a,\delta)$ is correct outside of $\delta \in \left[1-\frac{1}{\sqrt{2}}, 1-\frac{1}{\sqrt{2}}\right]$: without an internal minimum point, $GH\delta(a,h,\delta)$ from $h = 0$ to $h = \frac{1}{\sqrt{2}}$ is monotonically increasing when $\delta < 0$ and monotonically decreasing when $\delta > 0$.

We'll next justify our definition of $\omega\delta(a,\delta)$ for $\delta \in \left[1-\frac{1}{\sqrt{2}}, 1-\frac{1}{\sqrt{2}}\right]$.

18.1 We now first turn our attention[24] to showing that the derivative of $\omega\delta(a,\delta)$ is <u>positive</u> for $a > 1$ and $\delta \in \left[0, 1-\frac{1}{\sqrt{2}}\right]$.

Over the domain $0 \le h \le \frac{1}{\sqrt{2}}$, $GH\delta\left(a,\frac{1}{\sqrt{2}},\delta\right) \le GH\delta(a,0,\delta)$ by Lemma 10 and $GH\delta(a,h_K,\delta)$ is a minimum by Lemma 11. So we conclude that for $\delta \in \left[0, 1-\frac{1}{\sqrt{2}}\right]$, $\omega\delta(a,\delta) = GH\delta(a,0,\delta) - GH\delta(a,h_K,\delta)$

From Lemma 13, $GH\delta(a,h_K,\delta) = \sqrt{2}\left|a+\frac{\delta}{2}-\frac{2+\sqrt{2}}{4}\right|$. However, since $a > 1$ and $\delta \ge \frac{1}{\sqrt{2}} - a$, we only need to consider the case that $a+\frac{\delta}{2}-\frac{2+\sqrt{2}}{4} > 0$. Likewise, from the proof of Lemma 8, $a > 1$ is a sufficient condition for $GH\delta(a,0,\delta)$ to be positive.

[24] Credit: Algebra and derivatives calculations, and proof design assisted by Google Gemini, Google, August 26, 27 and September 3, 12-15, and 21, 2026. https://www.google.com/ In particular, Gemini suggested the idea of rewriting the cases as equalities with a *k* term. Algebra calculations assisted by Wolfram Alpha LLC. 2026. Wolfram|Alpha. http://www.wolframalpha.com (access September 12, 2026).

We take the partial derivative of $\omega\delta(a,\delta)$ with respect to δ and write it over a common denominator:

$$\frac{\partial}{\partial\delta}\omega(a,\delta)=\frac{a+\delta-\frac{\sqrt{2}}{2}-GH\delta(a,0,\delta)\left(\frac{\sqrt{2}}{2}\right)}{GH\delta(a,0,\delta)}$$

We know that $GH\delta(a,0,\delta)$ is positive, so to show the derivative to be non-negative, we merely need to show that the numerator is non-negative. To find the sign of the numerator, let $Y=GH\delta(a,0,\delta)\left(\frac{\sqrt{2}}{2}\right)$ and let $X=a+\delta-\frac{\sqrt{2}}{2}$. Then the numerator is $X-Y$

To calculate *X*, recall that $a+\frac{\delta}{2}-\frac{2+\sqrt{2}}{4}>0$. So, by definition of ">" we know $a+\frac{\delta}{2}-\frac{2+\sqrt{2}}{4}-k=0$ for some $k>0$ and so $a=-\frac{\delta}{2}+\frac{2+\sqrt{2}}{4}+k$. Substituting into *X* we get $X=a+\delta-\frac{\sqrt{2}}{2}=-\frac{\delta}{2}+\frac{2+\sqrt{2}}{4}+k+\delta-\frac{\sqrt{2}}{2}=\frac{\delta}{2}+\frac{2+\sqrt{2}}{4}+k-\frac{\sqrt{2}}{2}$. *X* is at its minimum when when $\delta=0$, where $X=\frac{2-\sqrt{2}}{4}+k$, which is positive.

Next, we use a difference of squares argument:

We can work out with some algebraic calculation that $X^2-Y^2=a\delta+\frac{\delta^2}{2}+\frac{2-\sqrt{2}}{2}a-\delta\frac{\sqrt{2}}{2}-\frac{1}{4}$

Substituting $a=-\frac{\delta}{2}+\frac{2+\sqrt{2}}{4}+k$ into X^2-Y^2 we find, after further algebraic calculation that $X^2-Y^2=k\delta+\frac{2-\sqrt{2}}{2}k$

Recall that $k>0$ and $\delta\geq0$. Thus, both the first and second terms are at least zero and perhaps positive. So we conclude that $X\text{^}2-Y\text{^}2>0$, and thus $X\text{^}2>Y\text{^}2$. Taking square roots, we infer $|X|>|Y|$. But *X* and *Y* are known to be positive, so $X>Y$ and thus $X-Y>0$. The numerator $X-Y$, and the derivative as a whole, is positive. So, to conclude for $a>1$ and $\delta\in\left[0,1-\frac{1}{\sqrt{2}}\right]$ the derivative of $\omega\delta(a,\delta)$ is positive.

18.2 Then, we turn our attention[25] to justifying and characterizing the next piece of the function on the other side of the $\delta=0$ boundary: that the derivative of $\omega\delta(a,\delta)$ is <u>negative</u> for $a>1$ and $\delta\in\left[\frac{1}{\sqrt{2}}-1,0\right]$.

Over the domain $0\leq h\leq\frac{1}{\sqrt{2}}$, $GH\delta\left(a,\frac{1}{\sqrt{2}},\delta\right)\geq GH\delta(a,0,\delta)$ by Corollary 2 and $GH\delta(a,h_K,\delta)$ is a minimum by Lemma 11. This is true even in the case where $GH\delta(a,0,\delta)=GH\delta(a,h_K,\delta)$, which occurs when $\delta=\frac{\sqrt{2}}{2}-1$, as noted in Lemma 11. So we know that for $\delta\in\left[\frac{1}{\sqrt{2}}-1,0\right]$, $\omega\delta(a,\delta)=GH\delta\left(a,\frac{1}{\sqrt{2}},\delta\right)-GH\delta(a,h_K,\delta)$.

As above, since $a>1$ and $\delta\geq\frac{1}{\sqrt{2}}-a$, we only need to consider the case when $a+\frac{\delta}{2}-\frac{2+\sqrt{2}}{4}>0$. We also have from the proof of Lemma 9 that $a>1$ is a sufficient condition for $GH\delta\left(a,\frac{1}{\sqrt{2}},\delta\right)$ to be positive.

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[25] Credit: Algebra and derivatives calculations, and proof design assisted by Google Gemini, Google, August 26&27 and September 3,12-15, and 21, 2026. https://www.google.com/ In particular, Gemini suggested the idea of rewriting inequalities as equalities with a *k* term and attempting to find the sign of *X*. Algebra calculations assisted by Wolfram Alpha LLC. 2026. Wolfram|Alpha. http://www.wolframalpha.com (access September 13, 2026).

We take the partial derivative of $\omega\delta(a,\delta)$ with respect to δ and write it over a common denominator:

$$\frac{\partial}{\partial\delta}\omega(a,\delta)=\frac{a+\delta-1-GH\delta\left(a,\frac{1}{\sqrt{2}},\delta\right)\left(\frac{\sqrt{2}}{2}\right)}{GH\delta\left(a,\frac{1}{\sqrt{2}},\delta\right)}$$

Since we know $GH\delta\left(a,\frac{1}{\sqrt{2}},\delta\right)$ is positive, the sign of the derivative as a whole depends on the sign of the numerator. To find the sign of the numerator, let $X=a+\delta-1$ and let $Y=GH\delta\left(a,\frac{1}{\sqrt{2}},\delta\right)\left(\frac{\sqrt{2}}{2}\right)$ $Y>0$ since *Y* represents a positive constant multiplying the known positive $GH\delta\left(a,\frac{1}{\sqrt{2}},\delta\right)$.

Then the numerator is $X-Y$. Now $a+\frac{\delta}{2}-\frac{2+\sqrt{2}}{4}>0$, so by definition of ">" we know $a+\frac{\delta}{2}-\frac{2+\sqrt{2}}{4}-k=0$ for some $k>0$ and so $a=-\frac{\delta}{2}+\frac{2+\sqrt{2}}{4}+k$. But in this specific aspect of $\omega\delta(a,\delta)$ we can't use this information to tell anything about sign of *X*.

Once again, we use a difference of squares argument.

We can work out with some algebraic calculation that $X^2-Y^2=a\delta+\frac{\delta^2}{2}+\frac{-2+\sqrt{2}}{2}a-\delta+\frac{1}{4}$

Substituting $a=-\frac{\delta}{2}+\frac{2+\sqrt{2}}{4}+k$ into X^2-Y^2 we find, after further algebraic calculation that $X^2-Y^2=k\delta+\frac{-2+\sqrt{2}}{2}k$

As for the first term, since $k>0$ and $\delta\leq 0$, $k\delta$ is at most 0. Likewise for the second term, $\frac{-2+\sqrt{2}}{2}$ is negative so when multiplied by *k*, the second term is negative. Thus X^2-Y^2 is negative, i.e. $X^2-Y^2<0$

Taking square roots, we have $|X|<Y$ since we know *Y* is positive. Since for all *X*, $X\leq|X|$, we can see by transitivity that $X\leq|X|<Y$ and $X<Y$ and so $X-Y<0$. The numerator $X-Y$, and the derivative as a whole, is negative. So, to conclude for $a>1$ and $\delta\in\left[1-\frac{1}{\sqrt{2}},0\right]$ the derivative of $\omega\delta(a,\delta)$ is negative.

To summarize the proof sketch so far: we've been able to show that the derivative of $\omega\delta(a,\delta)$ is negative for $a>1$ and $\delta\in\left[1-\frac{1}{\sqrt{2}},0\right]$ and positive for $a>1$ and $\delta\in\left[0,1-\frac{1}{\sqrt{2}}\right]$.

18.3 Now we turn our attention[26] to showing that the derivative of $\omega\delta(a,\delta)=GH\delta\left(a,\frac{1}{\sqrt{2}},\delta\right)-GH\delta(a,0,\delta)$ is <u>negative</u> for $a>1$ when $\frac{\sqrt{2}}{2}-a<\delta<\frac{\sqrt{2}}{2}-1$. (For our analysis, we do not need the derivative to be negative exactly at $\delta=\frac{\sqrt{2}}{2}-a$ even though this δ value is included in this piece of $\omega\delta(a,\delta)$.)

[26] Credit: Algebra calculations, and proof design assisted by Google Gemini, Google, August 26, 27 and September 16-19, 2026. https://www.google.com/

From our analysis above in 18.1 and 18.2, we can notice that the derivative of $GH\delta\left(a,\frac{1}{\sqrt{2}},\delta\right) - GH\delta(a,0,\delta)$:

$$\frac{\partial}{\partial\delta}\omega(a,\delta) = \frac{a+\delta-1}{GH\delta\left(a,\frac{1}{\sqrt{2}},\delta\right)} - \frac{a+\delta-\frac{\sqrt{2}}{2}}{GH\delta(a,0,\delta)}$$

By the remark at Corollary 2, and since $\delta < 0$ and $a > 1$, we know that

$$GH\delta\left(a,\frac{1}{\sqrt{2}},\delta\right) > GH\delta(a,0,\delta) > 0$$

Now, when $\frac{\sqrt{2}}{2} - a < \delta$, this implies that $0 < a + \delta - \frac{\sqrt{2}}{2}$, that is $\left(a + \delta - \frac{\sqrt{2}}{2}\right)$ is strictly positive.

Suppose $(a + \delta - 1) = 0$. Then $\frac{a+\delta-1}{GH\delta\left(a,\frac{1}{\sqrt{2}},\delta\right)} - \frac{a+\delta-\frac{\sqrt{2}}{2}}{GH\delta(a,0,\delta)}$ is negative since a positive value is being subtracted from zero.

Then, suppose $(a + \delta - 1) < 0$. Then $\frac{a+\delta-1}{GH\delta\left(a,\frac{1}{\sqrt{2}},\delta\right)} - \frac{a+\delta-\frac{\sqrt{2}}{2}}{GH\delta(a,0,\delta)}$ is negative since a positive value is being subtracted from a negative.

Finally, suppose $(a + \delta - 1) > 0$. From this we infer that $(a + \delta - 1) - \left(a + \delta - \frac{\sqrt{2}}{2}\right) = -1 + \frac{\sqrt{2}}{2} < 0$, which means $(a + \delta - 1) < \left(a + \delta - \frac{\sqrt{2}}{2}\right)$. Since $(a + \delta - 1)$ is positive and less than $(a + \delta - \sqrt{2}/2)$, Then

$$\frac{a+\delta-1}{GH\delta(a,0,\delta)} < \frac{a+\delta-\frac{\sqrt{2}}{2}}{GH\delta(a,0,\delta)}$$

And, because $GH\delta\left(a,\frac{1}{\sqrt{2}},\delta\right) > GH\delta(a,0,\delta) > 0$ and both numerators are positive,

$$\frac{a+\delta-1}{GH\delta\left(a,\frac{1}{\sqrt{2}},\delta\right)} < \frac{a+\delta-1}{GH\delta(a,0,\delta)}$$

then by transitivity,

$$\frac{a+\delta-1}{GH\delta\left(a,\frac{1}{\sqrt{2}},\delta\right)} < \frac{a+\delta-\frac{\sqrt{2}}{2}}{GH\delta(a,0,\delta)}$$

We conclude that $\frac{a+\delta-1}{GH\delta\left(a,\frac{1}{\sqrt{2}},\delta\right)} - \frac{a+\delta-\frac{\sqrt{2}}{2}}{GH\delta(a,0,\delta)} < 0$ when $(a + \delta - 1) > 0$.

So, the derivative of $\omega\delta(a,\delta)$ is negative when $a > 1$ and $\frac{\sqrt{2}}{2} - a < \delta$.

18.4 Now we turn our attention[27] to showing that the derivative of $\omega\delta(a,\delta) = GH\delta(a,0,\delta) - GH\delta\left(a,\frac{1}{\sqrt{2}},\delta\right)$ is <u>positive</u> for $a > 1$ when $\delta > 1 - \frac{\sqrt{2}}{2}$.

From our analysis above in 18.1 and 18.2, we can notice that the derivative of $GH\delta(a,0,\delta) - GH\delta\left(a,\frac{1}{\sqrt{2}},\delta\right)$:

$$\frac{\partial}{\partial\delta}\omega(a,\delta) = \frac{a+\delta-\frac{\sqrt{2}}{2}}{GH\delta(a,0,\delta)} - \frac{a+\delta-1}{GH\delta\left(a,\frac{1}{\sqrt{2}},\delta\right)}$$

Thus, we want to show that

$$\frac{a+\delta-\frac{\sqrt{2}}{2}}{GH\delta(a,0,\delta)} - \frac{a+\delta-1}{GH\delta\left(a,\frac{1}{\sqrt{2}},\delta\right)} > 0$$

when $\delta > 1 - \frac{1}{\sqrt{2}}$ and $a > 1$. From the discussion above, we know that when $a > 1$ that both $GH\delta\left(a,\frac{1}{\sqrt{2}},\delta\right)$ and $GH\delta(a,0,\delta)$ are positive. We also know that for $a > 1$ when $\delta > 1 - \frac{\sqrt{2}}{2}$ that $(a+\delta-1)$ is positive, and so the even larger $\left(a+\delta-\frac{\sqrt{2}}{2}\right)$ is also positive.

We begin by constructing *D* as a difference of squares

$$D = \left(\left(a+\delta-\tfrac{\sqrt{2}}{2}\right)GH\delta\left(a,\tfrac{1}{\sqrt{2}},\delta\right)\right)^2 - \left((a+\delta-1)GH\delta(a,0,\delta)\right)^2$$

When we multiply this out, we find that the result is $D = \delta^2 C_2 + \delta C_1 + C_0$

Where $C_0 = -2\sqrt{2}a^3 + 4a^3 - 3a^2 - a\sqrt{2} + 4a - \frac{3}{4}$, $C_1 = -3\sqrt{2}a^2 + 6a^2 - 3a + 2 - \frac{1}{\sqrt{2}}$, and $C_2 = -a\sqrt{2} + 2a - \frac{1}{2}$

D is positive since $\delta > 0$ and each of the coefficient functions C_2, C_1, and C_0 are already positive at $a = 1$ and their derivatives are strictly positive for all $a \geq 1$.

Since *D* is positive and both $\left(a+\delta-\frac{\sqrt{2}}{2}\right)GH\delta\left(a,\frac{1}{\sqrt{2}},\delta\right)$ and $(a+\delta-1)GH\delta(a,0,\delta)$ are positive when $\delta > 1 - \frac{1}{\sqrt{2}}$ and $a > 1$. we can conclude that $\left(a+\delta-\frac{\sqrt{2}}{2}\right)GH\delta\left(a,\frac{1}{\sqrt{2}},\delta\right) - (a+\delta-1)GH\delta(a,0,\delta) > 0$ and hence $\left(a+\delta-\frac{\sqrt{2}}{2}\right)GH\delta\left(a,\frac{1}{\sqrt{2}},\delta\right) > (a+\delta-1)GH\delta(a,0,\delta)$.

Then, by cross-dividing, which is proper and leaves the direction of the inequality unchanged since $GH\delta\left(a,\frac{1}{\sqrt{2}},\delta\right)$ and $GH\delta(a,0,\delta)$ are both positive when $a > 2$, we can write

[27] Credit: Algebra, limit calculation, and proof design assisted by Google Gemini, Google, September 17-19, 2026. https://www.google.com/ Algebra calculations assisted by Wolfram Alpha LLC. 2026. Wolfram|Alpha. http://www.wolframalpha.com (access September 18, 2026).

$$\frac{a+\delta-\frac{\sqrt{2}}{2}}{GH\delta(a,0,\delta)} > \frac{a+\delta-1}{GH\delta\left(a,\frac{1}{\sqrt{2}},\delta\right)}$$

which shows that $\frac{a+\delta-\frac{\sqrt{2}}{2}}{GH\delta(a,0,\delta)} - \frac{a+\delta-1}{GH\delta\left(a,\frac{1}{\sqrt{2}},\delta\right)} > 0$

18.5 We can now put all the pieces together and use[28] the Mean Value Theorem to find that any two-facet corner, where $a > 1$, the minimum possible parametrical error $\omega(a)$ occurs exactly when the corner is designed with a *d* value of $a - \frac{1}{\sqrt{2}}$.

Over the course of 18.1 through 18.4 we have justified and characterized the definition of $\omega\delta(a,\delta)$:

$$\omega\delta(a,\delta) = \begin{cases} GH\delta(a,0,\delta) - GH\delta(a,h_K,\delta) & \text{when } 1-\frac{\sqrt{2}}{2} \geq \delta \geq 0 & \frac{\partial}{\partial\delta} > 0 \\ GH\delta\left(a,\frac{1}{\sqrt{2}},\delta\right) - GH\delta(a,h_K,\delta) & \text{when } \frac{\sqrt{2}}{2}-1 \leq \delta \leq 0 & \frac{\partial}{\partial\delta} < 0 \\ GH\delta\left(a,\frac{1}{\sqrt{2}},\delta\right) - GH\delta(a,0,\delta) & \text{when } \frac{\sqrt{2}}{2}-a \leq \delta < \frac{\sqrt{2}}{2}-1 & \frac{\partial}{\partial\delta} < 0 \text{ for } \frac{\sqrt{2}}{2}-a < \delta < \frac{\sqrt{2}}{2}-1 \\ GH\delta(a,0,\delta) - GH\delta\left(a,\frac{1}{\sqrt{2}},\delta\right) & \text{when } \delta > 1-\frac{\sqrt{2}}{2} & \frac{\partial}{\partial\delta} > 0 \end{cases}$$

The Mean Value Theorem, to be applied, requires conditions of continuity and differentiability.

First, we can notice that each of the cases of $\omega\delta(a,\delta)$ is internally continuous over its domain, being the composition of roots of positive polynomial radicands and polynomials, all since $a > 1$.

Regarding continuity across the boundaries of the pieces of $\omega\delta(a,\delta)$, can find that $\omega\delta(a,\delta)$ is continuous at $\delta = 0$ because both the limit from the left of $GH\delta(a,0,\delta) - GH\delta(a,h_K,\delta)$ and the limit from the right of $GH\delta\left(a,\frac{1}{\sqrt{2}},\delta\right) - GH\delta(a,h_K,\delta)$ are equal to each other (viz. Corollary 3) and to the value of both pieces at $\delta = 0$

As for continuity across the other boundaries: At $\delta = \frac{1}{\sqrt{2}} - 1$, the limit from the left of $GH\delta\left(a,\frac{1}{\sqrt{2}},\delta\right) - GH\delta(a,0,\delta)$ equals the value and limit from the right of $GH\delta\left(a,\frac{1}{\sqrt{2}},\delta\right) - GH\delta(a,h_k,\delta)$. At $\delta = 1-\frac{1}{\sqrt{2}}$, the limit from the right of $GH\delta(a,0,\delta) - GH\delta\left(a,\frac{1}{\sqrt{2}},\delta\right)$ equals the value and limit from the left of $GH\delta(a,0,\delta) - GH\delta(a,h_k,\delta)$.

Thus, we can notice that all the pieces of the function $\omega\delta(a,\delta)$ can combine to form a continuous function over $\delta \in (-\infty,\infty)$.

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[28] Credit: Algebra calculations, differentiability, and proof design assistance by Google Gemini, Google, August 26, 27, September 16-18, and 21, 2026. https://www.google.com/

Regarding differentiability, we can notice that each of the cases of $\omega\delta(a,\delta)$ is internally continuous over its domain, being the composition of roots of positive polynomial radicands and polynomials, all since $a > 1$. We can see from the overlapping domains and derivatives at $\delta = 0$ that $\omega\delta(a,\delta)$ is not differentiable at $\delta = 0$ – and our analysis does not require it to be.

As for differentiability across the boundaries of the pieces of $\omega\delta(a,\delta)$, keeping in mind the constraint that $a > 1$: At $\delta = \frac{1}{\sqrt{2}} - 1$, the limit from the left of the partial derivative of $GH\delta\left(a,\frac{1}{\sqrt{2}},\delta\right) - GH\delta(a,0,\delta)$ equals the value and limit from the right of the partial derivative of $GH\delta\left(a,\frac{1}{\sqrt{2}},\delta\right) - GH\delta(a,h_k,\delta)$. At $\delta = 1 - \frac{1}{\sqrt{2}}$, the limit from the right of the partial derivative of $GH\delta(a,0,\delta) - GH\delta\left(a,\frac{1}{\sqrt{2}},\delta\right)$ equals the value and limit from the left of the partial derivative of $GH\delta(a,0,\delta) - GH\delta(a,h_k,\delta)$.

We thus see that $\omega\delta(a,\delta)$ has two disjoint regions of differentiability: $(0,\infty)$ and $(-\infty,0)$.

So, we can first infer that when $a > 1$ and $\delta > 0$ that $\omega\delta(a,\delta)$ is strictly increasing, so $\omega\delta_\delta(a,\delta)$ is strictly positive.

Now, $\omega\delta(a,\delta)$ is continuous in δ on $[0,\delta]$ and differentiable in δ on $(0,\delta)$ when $\delta > 0$. So by the Mean Value Theorem, there is a value c such that $0 < c < \delta$, such that $\omega\delta_\delta(a,c) = \frac{\omega\delta(a,\delta)-\omega\delta(a,0)}{\delta}$. Since both $\omega\delta_\delta(a,c)$ (i.e. rows 1 and 4 of the table above) and δ are positive, we conclude that $\omega\delta(a,\delta) - \omega\delta(a,0)$ must be positive, showing that when $\delta > 0$, $\omega\delta(a,\delta) > \omega\delta(a,0)$.

Next, we can infer that when $a > 1$ and $\frac{\sqrt{2}}{2} - a \le \delta < 0$ (since the geometric constraint $d \ge 0$ implies $\frac{\sqrt{2}}{2} - a \le \delta$) that $\omega\delta(a,\delta)$ is strictly decreasing, so $\omega\delta_\delta(a,\delta)$ is strictly negative.

Now, $\omega\delta(a,\delta)$ is continuous in δ on $[\delta,0]$ and differentiable in δ on $(\delta,0)$, where $\frac{\sqrt{2}}{2} - a \le \delta < 0$. We know that $\delta < 0$ because $\frac{\sqrt{2}}{2} - a < 0$ when $a > 1$. So, by the Mean Value Theorem, there is a value c such that $\delta < c < 0$ such that $\omega\delta_\delta(a,c) = \frac{\omega\delta(a,0)-\omega\delta(a,\delta)}{0-\delta}$. We know that $\omega\delta_\delta(a,c)$ is negative (i.e. rows 2 and 3 of the table above) for any such c, since c cannot be exactly $\frac{\sqrt{2}}{2} - a$ even if $\delta = \frac{\sqrt{2}}{2} - a$. Therefore, we conclude that since $0 - \delta$ is positive, then $\omega\delta(a,0) - \omega\delta(a,\delta)$ must be negative, showing that when $\frac{\sqrt{2}}{2} - a \le \delta < 0$, $\omega\delta(a,0) < \omega\delta(a,\delta)$

Because when $\delta > 0$, $\omega\delta(a,\delta) > \omega\delta(a,0)$ and when $\frac{\sqrt{2}}{2} - a \le \delta < 0$, $\omega\delta(a,\delta) > \omega\delta(a,0)$ can conclude that for all δ satisfying $\frac{\sqrt{2}}{2} - a \le \delta$, that is, for all δ that give rise to geometrically-meaningful non-negative d values, that $\omega\delta(a,0)$ is minimal. This minimality at $\delta = 0$ shows that $d = a - \frac{1}{\sqrt{2}} + \delta = a - \frac{1}{\sqrt{2}}$ gives the minimal parametrical error for all $a > 1$.

19. References

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20. Notices

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